Cancer Biology
Stem Cells and Regenerative Medicine

Reviewed Preprint
v1 • September 12, 2024
Not revised

Regulation of lung cancer initiation and progression by the stem cell determinant Musashi

Alison G Barber, Cynthia M Quintero, Michael Hamilton, Nirakar Rajbhandari, Roman Sasik, Yan Zhang, Carla F Kim, Hatim Husain, Xin Sun, Tannishtha Reya

Department of Pharmacology and Medicine, University of California San Diego School of Medicine, La Jolla, CA • Moores Cancer Center, University of California San Diego School of Medicine, La Jolla, CA • Herbert Irving Comprehensive Cancer Center, Columbia University Medical Center, New York City, NY • Department of Physiology and Cellular Biophysics, Columbia University Medical Center, New York City, NY • Center for Computational Biology and Bioinformatics, University of California San Diego School of Medicine, La Jolla, CA • Department of Pediatrics, University of California, San Diego, La Jolla, CA • Stem Cell Program, Division of Hematology/Oncology and Division of Respiratory Disease, Boston Children's Hospital, Boston, MA • Department of Genetics, Harvard Medical School, Boston, MA • Harvard Stem Cell Institute, Cambridge, MA

Abstract

Despite advances in therapeutic approaches, lung cancer remains the leading cause of cancer-related deaths. To understand the molecular programs underlying lung cancer initiation and maintenance, we focused on stem cell programs that are normally extinguished with differentiation but can be reactivated during oncogenesis. Here we have used extensive genetic modeling and patient derived xenografts to identify a dual role for Msi2: as a signal that acts initially to sensitize cells to transformation, and subsequently to drive tumor propagation. Using Msi reporter mice, we found that Msi2-expressing cells were marked by a pro-oncogenic landscape and a preferential ability to respond to Ras and p53 mutations. Consistent with this, genetic deletion of Msi2 in an autochthonous Ras/p53 driven lung cancer model resulted in a marked reduction of tumor burden, delayed progression, and a doubling of median survival. Additionally, this dependency was conserved in human disease as inhibition of Msi2 impaired tumor growth in patient-derived xenografts. Mechanistically, Msi2 triggered a broad range of pathways critical for tumor growth, including several novel effectors of lung adenocarcinoma. Collectively, these findings reveal a critical role for Msi2 in aggressive lung adenocarcinoma, lend new insight into the biology of this disease, and identify potential new therapeutic targets.

eLife assessment

This **important** study shows a significant role for Mushashi-2 (Msi2) in lung adenocarcinoma. The authors provided **solid** data that support the requirement for Msi2 in tumor growth and progression, although the study would have been strengthened by including more patient samples and additional evidence regarding Msi2+ cells being more responsive to transformation. These findings are of interest to both the lung cancer and the RNA binding protein fields.

https://doi.org/10.7554/eLife.97021.1.sa4

Introduction

Lung cancer is the second most common type of cancer. While the death rate due to lung cancer has decreased over time due to a drop in the prevalence of smoking, it is still by far the leading cause of cancer-related deaths in men and women, resulting in more deaths than colon, breast, and prostate cancer combined[1]. Non-small-cell lung cancer (NSCLC) is the most prevalent form of lung cancer, accounting for 84% of lung cancer cases. Within this cancer type, adenocarcinoma is the most frequent subtype, accounting for 50% of all NSCLCs[2,3]. Most cases of lung cancer are diagnosed at late-stage— when there is regional or distant spread of the disease—at which time treatment options are limited[4]. Despite advances in lung cancer therapy, the 5-year survival rates for late-stage (i.e. regional or distant) NSCLC are 37% to 9%, respectively[5]. These survival rates highlight the critical need for improved understanding of the biology of this disease in order to develop more effective treatments.

To understand the key dependencies of lung cancer, we have focused on stem cell programs which are often co-opted to perpetuate an aggressive, undifferentiated state. The stem cell fate determinant Musashi-2 (Msi2) is one such signal that has been shown to be critical in development as well as advanced cancers[6,7,8]; however, its role in lung cancer is not well understood. Msi2 has been reported to be overexpressed in human lung adenocarcinoma and regional metastases, and its inhibition in human cell lines has been shown to reduce invasive and metastatic potential *in vitro*[9,10]. However, this study focused on *in vitro* analysis utilizing human NSCLC cell lines; thus, no evidence for a role of Msi2 *in vivo* using definitive genetic mouse models exist. Further, whether Msi2 is a dependency for tumor initiation or required for continued propagation, and which pathways are regulated by Msi2 to drive tumor growth, also remain unknown. Here we have utilized a combination of genetic models and patient-derived xenografts to determine the role of Msi2 signaling in the growth and progression of lung adenocarcinoma. Using Msi2 knock-in reporter mice and Msi2 conditional knockout mice, we show that Msi2 plays a dual role in lung adenocarcinoma, acting as a signal that not only is essential for initiation, but one that continues to be required post-establishment in both genetically-engineered mouse models and in patient-derived xenografts. Finally, using RNA-sequencing analysis we show that Msi2 can trigger a range of pathways critical for tumor growth, including several new effectors of lung adenocarcinoma. These findings suggest that Msi2 is a critical regulator of lung adenocarcinoma and offer new insight into the signals that facilitate transformation and support disease progression.

Results

Msi2 is expressed in normal lung and in adenocarcinoma

To determine whether Musashi regulates the growth and progression of aggressive lung cancer we first determined whether Msi2 is expressed in stem cells of the distal lung. We used a $Msi2^{eGFP/+}$ knock-in reporter mouse to track endogenous Msi2 expression[6], and found that Msi2 is expressed in 39% of $Lin^{-}EpCAM^{+}$ lung epithelial cells (**Fig. 1A**) and, more specifically, in 37% of cells enriched for Club/BASC cells and 26% of cells enriched for AT2 cells—the known stem cell populations of the distal lung (**Fig. 1B-C**). Interestingly, although Msi2 expression could only be detected in a fraction of distal lung stem cells, widespread expression of Msi2 was often observed in lung adenocarcinomas formed in $Kras^{G12D/+}$; $p53^{fl/fl}$ mice (**Fig. 1D**). The expansion of Msi2 expression in lung tumors compared to normal lung epithelia suggests that normal cells expressing Msi2 are more sensitive to transformation, or that once transformed, Msi2 expression is amplified in tumor cells.

Msi2-expressing cells are more responsive to transformation by mutant Ras and p53

To determine whether Msi2 expression marks cells with increased capacity for tumor growth, we generated Msi2-reporter $Kras^{G12D/+}$; $p53^{fl/fl}$ mice with inducible Cre expression ($Msi2^{eGFP/+}$; $Kras^{G12D/+}$; $p53^{fl/fl}$; $RosaCre^{ER/+}$; **Fig. 2A**), delivered tamoxifen, isolated $Msi2^{+}$ and $Msi2^{-}$ lung epithelial cells, and tested sphere-forming capacity. As shown in **Fig. 2B**, $Msi2^{+}$ cells had enhanced sphere-forming capacity, with a 6-fold increase in sphere-forming observed at primary passage, and this capacity was retained for up to 4 passages (**Fig. 2B**). Further, spheres that were isolated and transplanted subcutaneously into the flanks of immunocompromised mice developed Msi2-expressing tumors *in vivo* as well as metastatic lesions in the lung, demonstrating that the observed sphere-forming capacity is reflective of tumorigenicity and not merely stem cell capacity (**Fig. 2C-E**). To understand the basis for the enhanced tumorigenic capacity of $Msi2^{+}$ cells, we carried out RNA-sequencing (RNAseq) to examine the transcriptional landscape of Msi2-expressing cells in normal lung epithelia. Interestingly, the differentially expressed gene (DEG) profile showed that $Msi2^{+}$ cells are transcriptionally distinct from $Msi2^{-}$ cells (**Fig. 2F**). Gene Set Enrichment Analysis (GSEA) further revealed that $Msi2^{+}$ cells are significantly enriched for signatures related to developmental and stem cell signaling, consistent with the known role of Msi2 as a critical regulator of the stem cell state[11,12,13], as well as signatures related to oncogenic signaling and therapy resistance (**Fig. 2G**, **2I**, **2K**) which are associated with aggressive cancers. Over 1,900 differentially expressed genes were found to be enriched in the $Msi2^{+}$ population. In line with the GSEA results, these enriched genes included those involved in developmental and stem cell signaling (such as *Dll1*, *Jag2*, and *Notch3)* [14–17], oncogenic signaling (such as *Akt3, Ret, Myb, and Fos*)[18–21], and therapy resistance (such as *Axl*, *Egr1*, and several glutathione S transferases implicated in platinum drug resistance including *Gstp1 and Gsta2*)[22,23,24] (**Fig. 2H**, **2J**, **2L**). These findings suggest that prior to tumor initiation, the transcriptional landscape of Msi2-expressing cells is enriched in signaling programs that are conducive to tumorigenesis, which may make them uniquely poised for transformation.

Genetic deletion of Msi2 leads to a decrease in tumor incidence, burden, and progression

To determine if Msi2 not only marks a cell with an enhanced capacity for lung adenocarcinoma growth, but whether it may also be required for initiation of these tumors, we compared wild type and Msi2 knock-out $Kras^{G12D/+}$; $p53^{fl/fl}$ mice ($Msi2^{-/-}$; $Kras^{G12D/+}$; $p53^{fl/fl}$) (**Fig. 3A**). In this model,

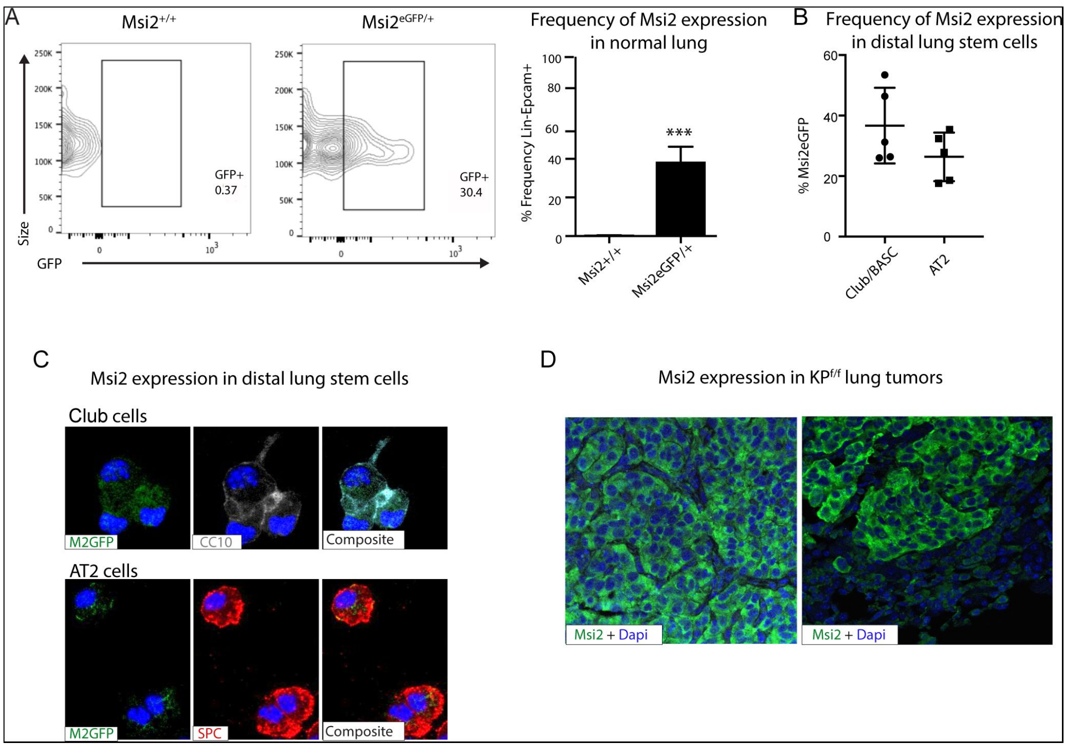


Figure 1

Msi2 is expressed in the stem cells of the distal lung and in lung adenocarcinoma.

A) Msi2 is expressed in the Lin$^-$EpCAM$^+$ lung epithelial cells of Msi2$^{GFP/+}$ (M2GFP) reporter mice. Representative FACS plots shown for non-reporter (Msi2$^{+/+}$, left) and M2GFP reporter mouse lungs (Msi2$^{eGFP/+}$middle). Frequency of M2GFP-expressing cells in lung epithelia (right; n = 2 non-reporter, n = 5 reporter). Data represented as mean ± SD, one outlier identified and removed using the Grubb's test, ***p < 0.001 by Student's t-test. B-C) Msi2 is expressed in the known stem cell populations of the distal lung. B) Frequency of Msi2 expression in the Lin$^-$EpCAM$^+$Sca1$^+$ enriched Club/BASC and the Lin$^-$EpCAM$^+$Sca1$^-$ enriched AT2 cell populations (n=5). C) Representative images of cytospins from M2GFP reporter mouse Lin$^-$EpCAM$^+$ lung epithelial cells showing expression of Msi2 expression in Club cells (top row, marked by CC10 staining) and AT2 cells (bottom row, marked by SPC staining). D) Representative images of Msi2 in tumors from KP$^{f/f}$ mice; some tumors display ubiquitous expression of Msi2 (left), while others display more heterogeneous expression (right).

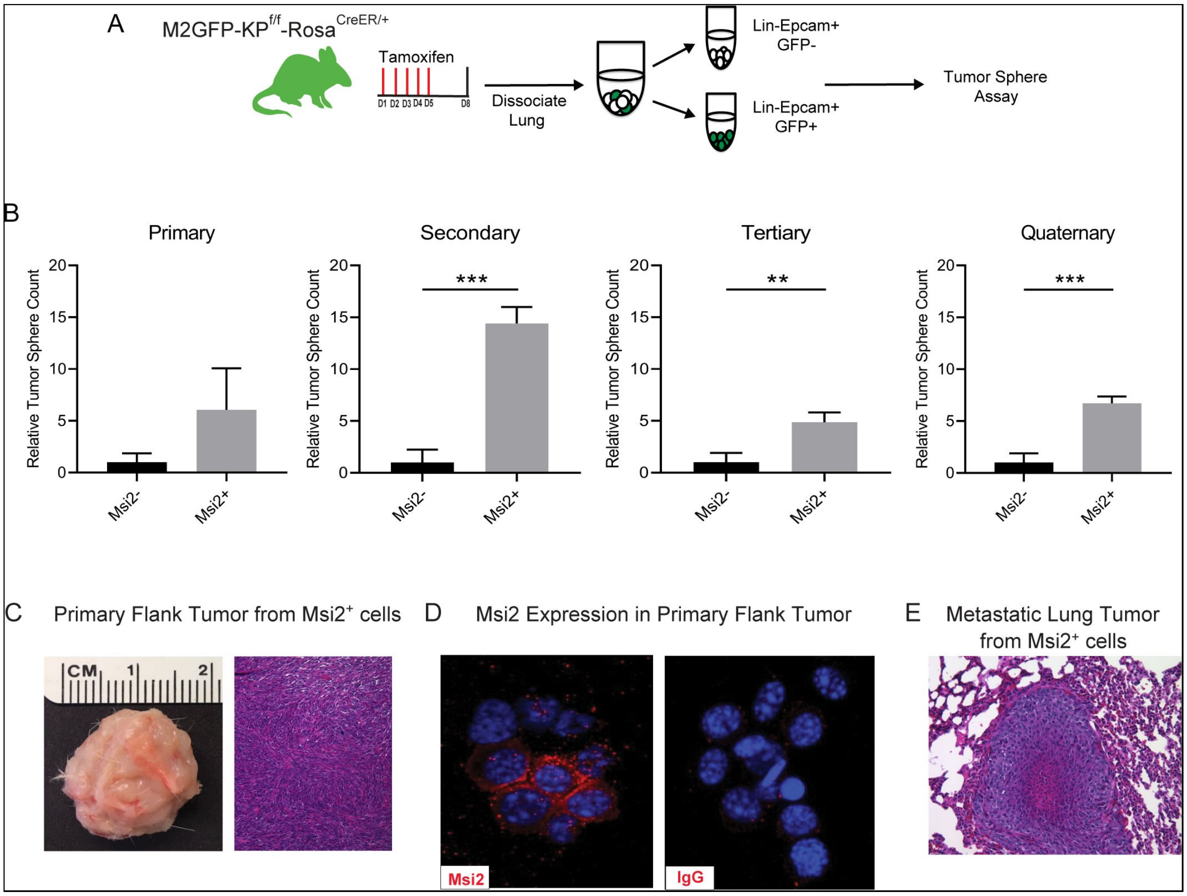


Figure 2

Msi2-expressing cells preferentially favor tumor growth in lung adenocarcinoma.

A) Schematic of the strategy used to measure the tumor sphere forming capacity of Msi2-expressing and non-expressing cells *in vitro*. Msi2GFP/+; KrasG12D/+; p53fl/fl; RosaCreER/+ (M2GFP-KPf/f -RosaCreER/+) mice were treated with tamoxifen for 5 days to induce recombination of floxed alleles. Three days following the final dose of tamoxifen, lungs were dissociated and Msi2-expressing (GFP+) or non-expressing (GFP-) cells were isolated by FACS and then plated in a tumor sphere assay. B) Msi2-expressing cells isolated from KPf/f lung epithelia following tamoxifen treatment preferentially form tumor spheres over multiple cell passages *in vitro* as compared to non-expressing cells. Representative experiment shown, (n = 3). Data represented as mean ± SD, $^{**}p<0.01$, $^{***}p<0.001$ by Student's t-test. C-E) Tumor spheres formed by Msi2+ cells *in vitro* form aggressive tumors *in vivo*. Tumor spheres isolated from Msi2+ cells after quaternary passage *in vitro* and transplanted *in vivo* form highly aggressive flank tumors (C) that retain Msi2 expression (D) and are able to metastasize to the lung (E). F) Heatmap showing significant differentially expressed genes between Msi2-expressing cells and non-expressing cells. G-L) Gene Set Enrichment Analysis of Msi2-expressing normal lung epithelial cell gene signatures. Heatmaps of gene signatures and selected genes that are involved in developmental and stem cell signaling (G-H), oncogenic signaling (I-J), and therapy resistance (K-L). Red represents gene signatures or genes that are upregulated in the presence of Msi2, while blue represents gene signatures or genes that are downregulated in the absence of Msi2.

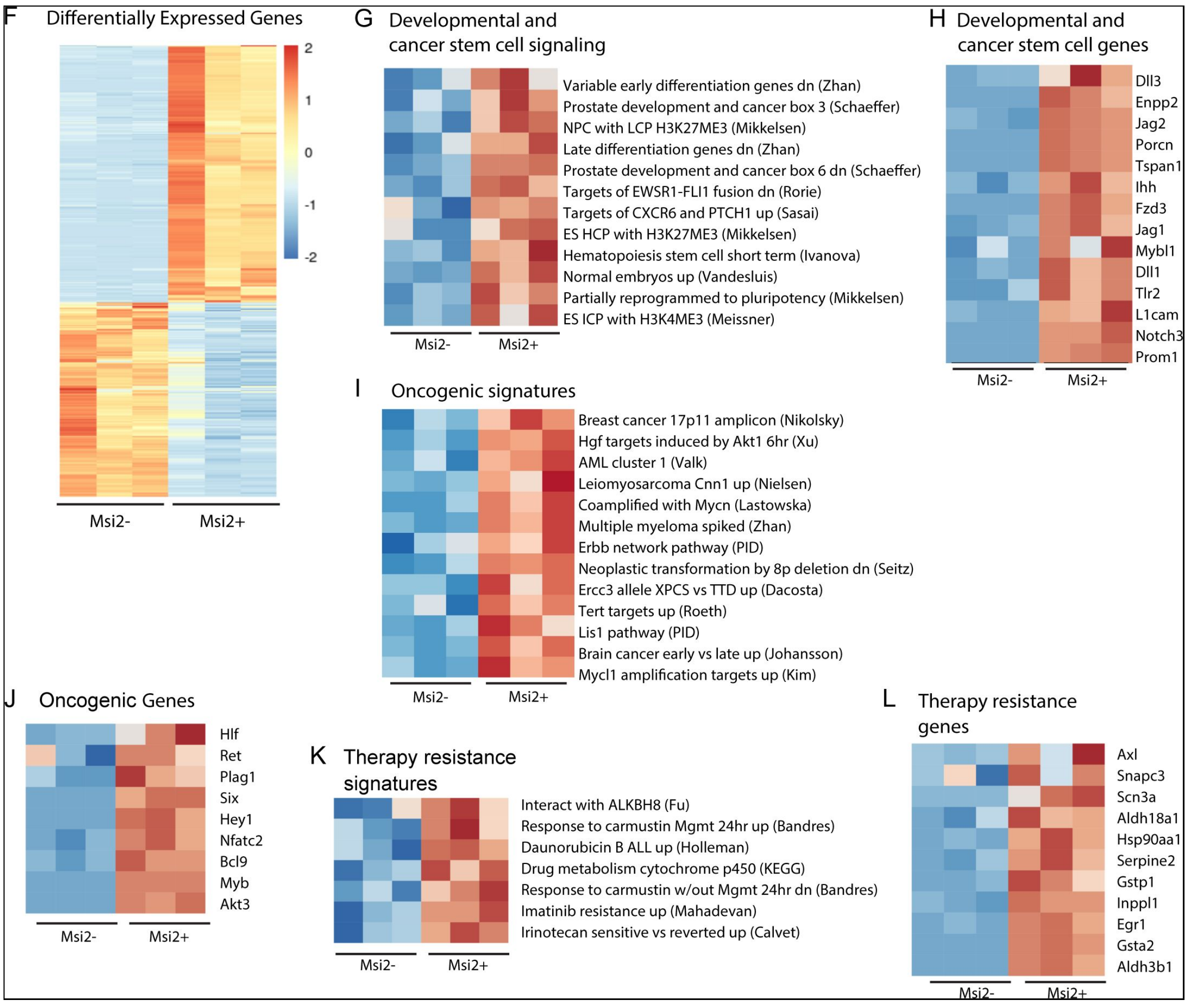


Figure 2 (continued)

the *Msi2* gene has been disrupted via gene trap mutagenesis[7], while the inhalation of adenoviral-Cre (Ad-Cre) results in the conditional activation of Kras and inactivation of p53 in the lung[25]. Importantly, loss of Msi2 led to the formation of significantly fewer (84%) tumor lesions as well as a significantly lower (85%) overall tumor burden in the lung at a 14-week midpoint (**Fig. 3B-D**). Further, while there was no difference in the frequency of low-grade tumors, there was a significant decrease in the frequency of mid- and high-grade tumors (45-54%), suggesting that loss of Msi2 delayed tumor progression from low to high grade (**Fig. 3E**). Importantly, Msi2$^{-/-}$ KP$^{f/f}$ mice infected with Ad-Cre developed tumors less frequently than their wild type counterparts (61% Msi2$^{-/-}$ vs 83% Msi2$^{+/+}$; **Fig. 3F**). To confirm that the population of lung stem cells is not reduced in Msi2 knockout mice and therefore contributing to a decrease in tumor lesions, we stained the lungs of WT and Msi2 knockout mice for Club/BASC and AT2 lung stem cells and saw no difference (data not shown). Additionally, Msi2$^{-/-}$ KP$^{f/f}$ mice displayed a significant increase in overall survival compared to Msi2$^{+/+}$ KP$^{f/f}$ mice. Msi2$^{+/+}$ KP$^{f/f}$ mice had a median survival of 209 days while the Msi2$^{-/-}$ KP$^{f/f}$ mice had a median survival of 428 days, reflecting a 2-fold increase in survival ($p = 0.04$, **Fig. 3G**). Notably, the majority of lung tumors that formed in Msi2$^{-/-}$ KP$^{f/f}$ mice were found to express Msi2, suggesting that the tumors that formed in Msi2$^{-/-}$ KP$^{f/f}$ mice were escapers that re-expressed Msi2, underscoring the dependency on Msi2 signaling for tumor formation (**Fig. 3H**). Taken together these results suggest that Msi2 not only marks cells with an enhanced capacity for tumor growth but is also required for initiation and progression of lung adenocarcinoma.

The analyses described above were performed in Msi2$^{-/-}$ KP$^{f/f}$ mice, and therefore in an environment in which Msi2 is absent at the onset of tumorigenesis. To test whether established cancers have a similar dependency on Msi2 for their growth, we knocked down Msi2 in primary tumor cell lines generated from Kras$^{G12D/+}$; p53$^{fl/fl}$ lung tumors (KP$^{f/f}$ cell line) and measured tumorsphere formation. Importantly, loss of Msi2 led to a significant decrease (75%) in tumorsphere formation *in vitro*, suggesting that Msi2 is required for tumor cell propagation (**Fig. 4A-B**). To confirm this finding *in vivo*, Msi2 knockdown KP$^{f/f}$ cells were transplanted into the flanks of syngeneic mice, and tumor growth monitored over time. As shown in **Figure 4**, the Msi2-knockdown tumors had significantly reduced (89%) tumor cell numbers (**Fig. 4C-D**), indicating that established tumors remain dependent on Msi2 signaling for growth. To determine the relevance of our findings for human disease, we examined the effect of Msi2 inhibition on primary patient samples. To this end, patient samples were transplanted into immunocompromised mice to generate patient-derived xenografts (PDXs). Once established, PDX tumors were harvested, dissociated, transduced with either shControl or shMsi2 lentivirus, and subsequently transplanted into the flanks of immunocompromised mice (**Fig. 4E**). Importantly, although an equivalent number of shControl and shMsi2 cells were transplanted into recipient mice, there was a notable delay in the growth of Msi2-knockdown tumors, which were 64-88% smaller than control tumors at endpoint (**Fig. 4F**, **Supplementary Fig. 1**). These findings suggest that human lung adenocarcinomas are dependent on Msi2 for their continued growth.

Msi2 regulates signaling pathways critical for tumor growth

To understand the mechanism by which Msi2 impacts tumor growth, we knocked down Msi2 in KP$^{f/f}$ cell lines and carried out an RNAseq analysis comparing control and knock-down cells. Principal component analysis shows a clear separation between the control and Msi2-knockdown cells and significant changes in over ~170 genes, highlighting the marked impact that loss of Msi2 has on the transcriptional profile of lung adenocarcinoma (**Fig. 5A**). Consistent with the role of Msi2 in stem cell maintenance, the inhibition of Msi2 in lung cancer cells had a significant impact on developmental and stem cell signaling programs and genes, this included *Porcn*, the transcription regulator *Nupr1*, and the NuRD complex member *Mbd3*, which have been shown to play a role in the maintenance of the stem cell state (**Fig. 5B-C**)[26–29]. In addition, Msi2 impacted a broad range of other signaling programs and their associated genes, including DNA repair and metabolism. DNA repair genes included *Brca1*, *Atm*, and several Fanconi anemia

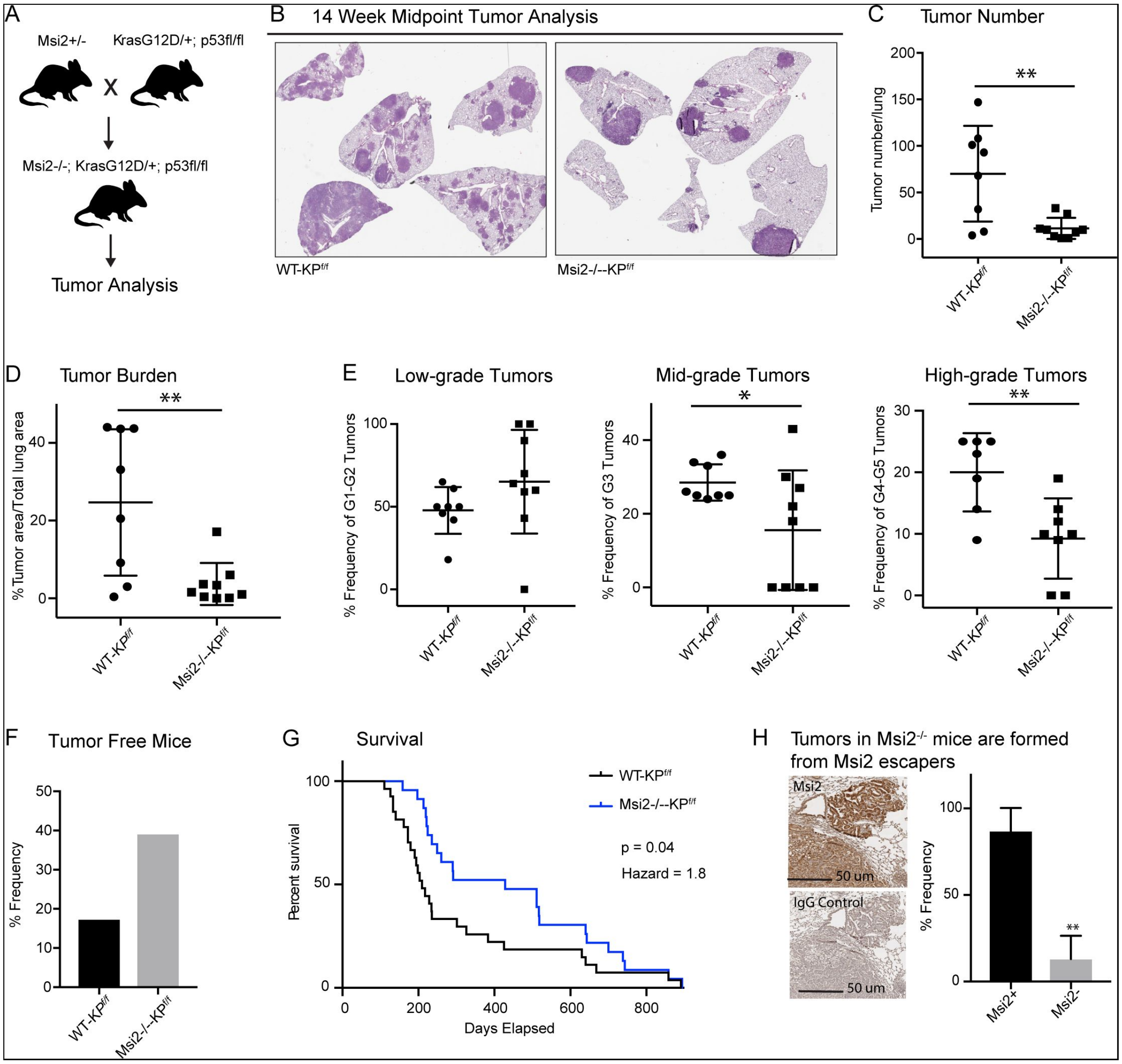


Figure 3

Msi2 is required for lung adenocarcinoma growth and progression.

A) Breeding scheme used to generate the $Msi2^{-/-}$; $Kras^{G12D/+}$; $p53^{fl/fl}$ ($Msi2^{-/-}$-$KP^{f/f}$) model of lung adenocarcinoma. B) Representative images of wild type (left) and $Msi2^{-/-}$ (right) $Kras^{G12D/+}$; $p53^{fl/fl}$ lungs gender and age matched mice at 14 weeks after tumor initiation. C-D) The number of tumors formed (C) and overall tumor burden (D) is significantly reduced in $Msi2^{-/-}$ $KP^{f/f}$ mice. E) $Msi2^{-/-}$ $KP^{f/f}$ mice have a significant reduction in the frequency of mid- and high-grade tumors. Data represented as mean ± SD, two outliers identified and removed for high-grade tumors using the Grubb's test *p < 0.05, **p<0.01 by Student's t-test. F) The tumor take rate is reduced in $Msi2^{-/-}$ $KP^{f/f}$ mice. While only 17% of AdCre-infected wild ype mice are tumor-free at the time of death, more than twice as many (39%) $Msi2^{-/-}$ $KP^{f/f}$ mice are tumor-ree at the time of death. G) $Msi2^{-/-}$ $KP^{f/f}$ have significantly increased survival (p = 0.04, hazard ratio = 1.8; n = 27 WT-$KP^{f/f}$, n = 23 $Msi2^{-/-}$-$KP^{f/f}$), with a median survival of 428 days compared to 209 days for wild type mice. Log-rank test was used to determine the difference in survival curves between wild type and $Msi2^{-/-}$ $KP^{f/f}$ mice. H) Lung tumors in $Msi2^{-/-}$ $KP^{f/f}$ mice express Msi2. Representative mmunohistochemical staining (left) for Msi2 (top) or IgG control (bottom) in lung tumors from a 16-week old $Msi2^{-/-}$ $KP^{f/f}$ mouse. Quantification of the frequency of Msi2-expressing tumors from $Msi2^{-/-}$ $KP^{f/f}$ mice shows hat while some tumors lack Msi2 expression the majority of tumors express Msi2 (right, n = 3). Data represented as mean ± SD, **p<0.01 by Student's t-test.

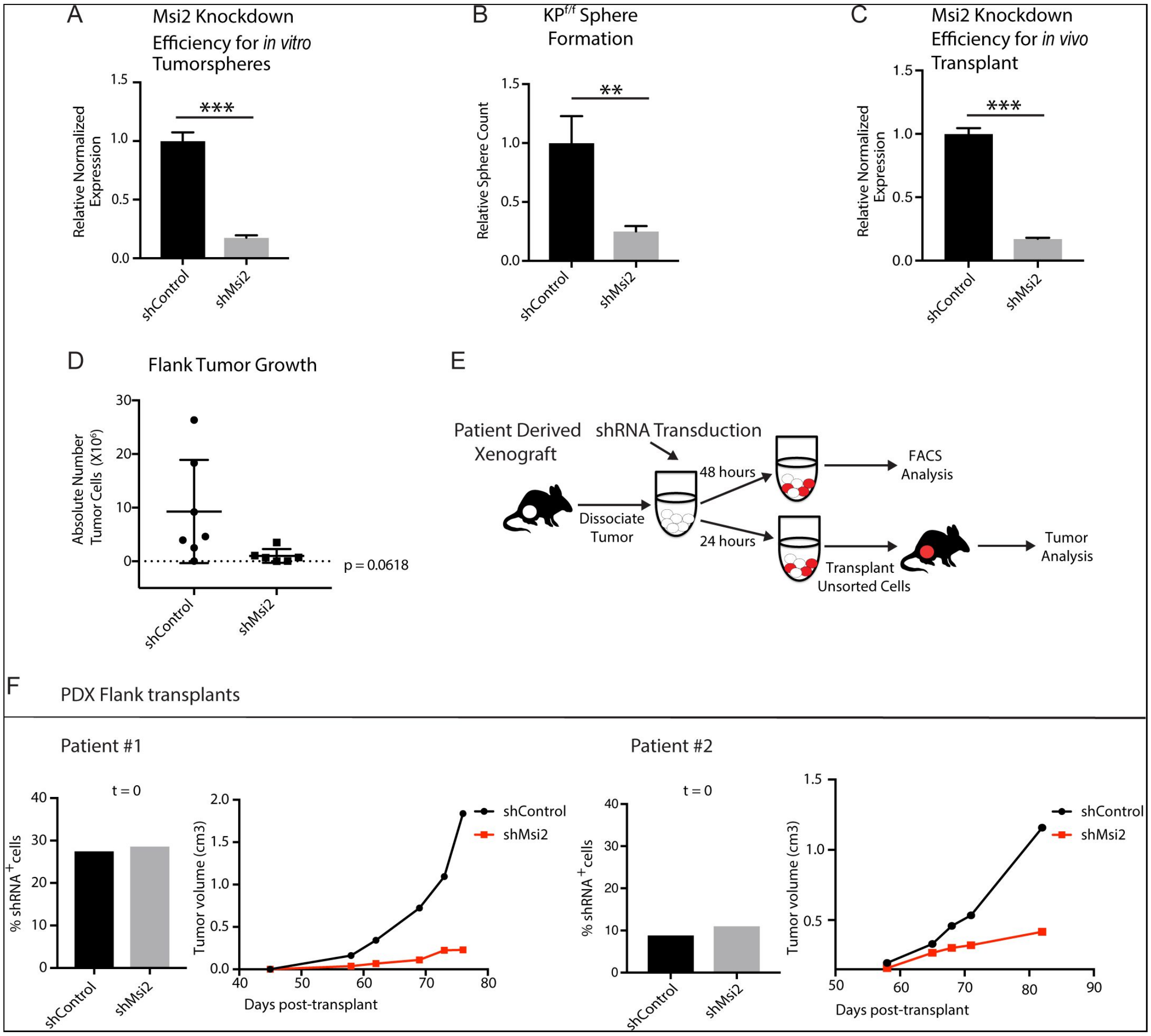


Figure 4

Loss of Msi2 impairs growth of established cancer.

A) Mouse $KP^{f/f}$ cells transduced with shMsi2 and used for *in vitro* tumorsphere assays have an 83% reduction in *Msi2* mRNA expression. B) Loss of *Msi2* significantly impairs tumor sphere formation *in vitro* (n =3). C) Mouse $KP^{f/f}$ cells transduced with shMsi2 and used for *in vivo* flank transplant assays have an 83% reduction in *Msi2* mRNA expression. D) Loss of *Msi2* consistently impairs tumor growth in flank transplants *in vivo*. Data represented as mean ± SD, one outlier identified and removed using the Grubb's test **p<0.01 by Student's t-test. E) Schematic for testing the impact of Msi2 loss on growth of patient- derived xenografts *in vivo*. Patient-derived xenografts were harvested, dissociated, transduced with shMsi2 or shControl virus and split into two pools that were incubated for either 24 or 48 hours. After 24 hours, one pool of cells (containing a mixture of transduced and untransduced cells) was transplanted into the flanks of NSG mice. After 48 hours the remaining pool of cells was analyzed via FACS to determine the frequency of cells infected with shRNA. F) The frequency of infection was comparable for shControl and shMsi2 in two independent patient samples. Reduced *Msi2* expression led to pronounced inhibition of tumor growth in two independent patient-derived xenografts.

complementation group genes[30,31,32], and metabolism programs included the gluconeogenesis enzyme *Pck2*, which has been shown to promote lung cancer cell survival in low glucose conditions, several isoforms of *Cpt1*, a key regulator in fatty acid oxidation known to promote tumor growth in a variety of cancers, and the glutamate dehydrogenase *Glud1*, which has been shown to promote tumorigenesis in lung cancer xenografts (**Fig. 5D**-G[33,34,35]). Interestingly, several known regulators of lung adenocarcinoma such as *Gli1*, a canonical downstream effector of the Hedgehog signaling pathway, βIII tubulin (*Tubb3)*, and CC Chemokine Receptor 1 (*Ccr1)* were found to be significantly downregulated following Msi2 inhibition as well (**Fig. 5H**)[36–44]. These data suggest that Msi2 is a regulator of a variety of signaling pathways that are critical to tumor growth and lung cancer progression. In addition to these known regulators, a number of genes not previously implicated in lung cancer were found to be significantly downregulated following Msi2 inhibition, suggesting that they may play a role in tumor growth (**Fig. 5I-J**). To identify potential novel regulators of lung adenocarcinoma we focused on significantly downregulated genes that were highly expressed, which narrowed our gene list to 30 potential candidates. PCR based validation and prioritization of those genes with no known role in lung cancer led to a subset that included genes such as the E3 ubiquitin ligase *Rnf157*, which promotes neuronal growth through the inhibition of apoptosis[45], Synaptotagmin 11 (*Syt11)*, a critical mediator of neuronal vesicular trafficking[46], prostoglandin D2 synthase (*Ptgds*) known to be involved in reproductive organ development[47], and ADP ribosylation like binding protein (*Arl2bp)*, a known regulator of STAT3 signaling[48]. To test whether these genes regulate tumor cell growth we knocked down each of these genes (*Ptgds*, *Arl2bp*, *Rnf157*, and *Syt11*) in the $KP^{f/f}$ cell line. Inhibition of each target led to a significant decrease (88-98%) in the formation of tumor spheres *in vitro* (**Fig. 5K**). These findings not only identify these genes as downstream effectors of Msi2 activity but more generally show that this dataset could be an important resource to identify new regulators of lung adenocarcinoma growth.

Discussion

The stem cell fate determinant Msi2 is commonly found to be expressed in carcinomas[10,49,50,51] and Msi signaling has been shown to contribute to colon adenomas[52] and maintain the aggressive, undifferentiated state in hematologic malignancies and pancreatic cancer[7,6]. Here we have used genetically-engineered and autochthonous models to show that Msi2 is critically required in the development and progression of primary lung adenocarcinoma. Msi2 signaling emerges as a key dependency at initiation and continues to be needed during tumor progression. Using the well-established $Kras^{G12D}$; $p53^{flox/flox}$ ($KP^{fl/fl}$) mouse model of lung adenocarcinoma[53], we created a Msi2 knock-out lung adenocarcinoma model and found that deletion of Msi2 at initiation significantly impaired tumor incidence, reduced tumor burden and progression, and cumulatively led to a doubling of median survival. Furthermore, loss of Msi2 impaired propagation of established $KP^{fl/fl}$ tumors *in vitro* and *in vivo*. In order to determine whether these findings were relevant for the human disease, we used patient-derived xenografts, which have proven to be a highly valuable tool for understanding the biology of human cancers and have been shown to faithfully predict therapeutic response in a majority of patients[54,55,56]. Inhibition of Msi2 in PDXs also led to impaired tumor growth *in vivo*, suggesting that dependence on Msi2 is conserved in human lung adenocarcinoma.

Our finding that the loss of Msi2 prior to transformation significantly impacts tumor initiation prompts the interesting question of whether Msi2 signaling can influence early tumor growth for lung adenocarcinoma. Previous use of the $KP^{fl/fl}$ model has illustrated how both the cellular and molecular context at the time of transformation influences subsequent cancer development[57]. Using cell-type specific Ad-Cre to drive the expression of either $Kras^{G12D}$ alone or $Kras^{G12D}$ in combination with $p53^{flox/flox}$, work by Sutherland et al.[57] has shown that the cell of origin for adenocarcinoma differs based on the genetic mutations used to drive the cancer. When $Kras^{G12D}$ alone was used as the driver, AT2 cells were able to form adenomas which then progressed to

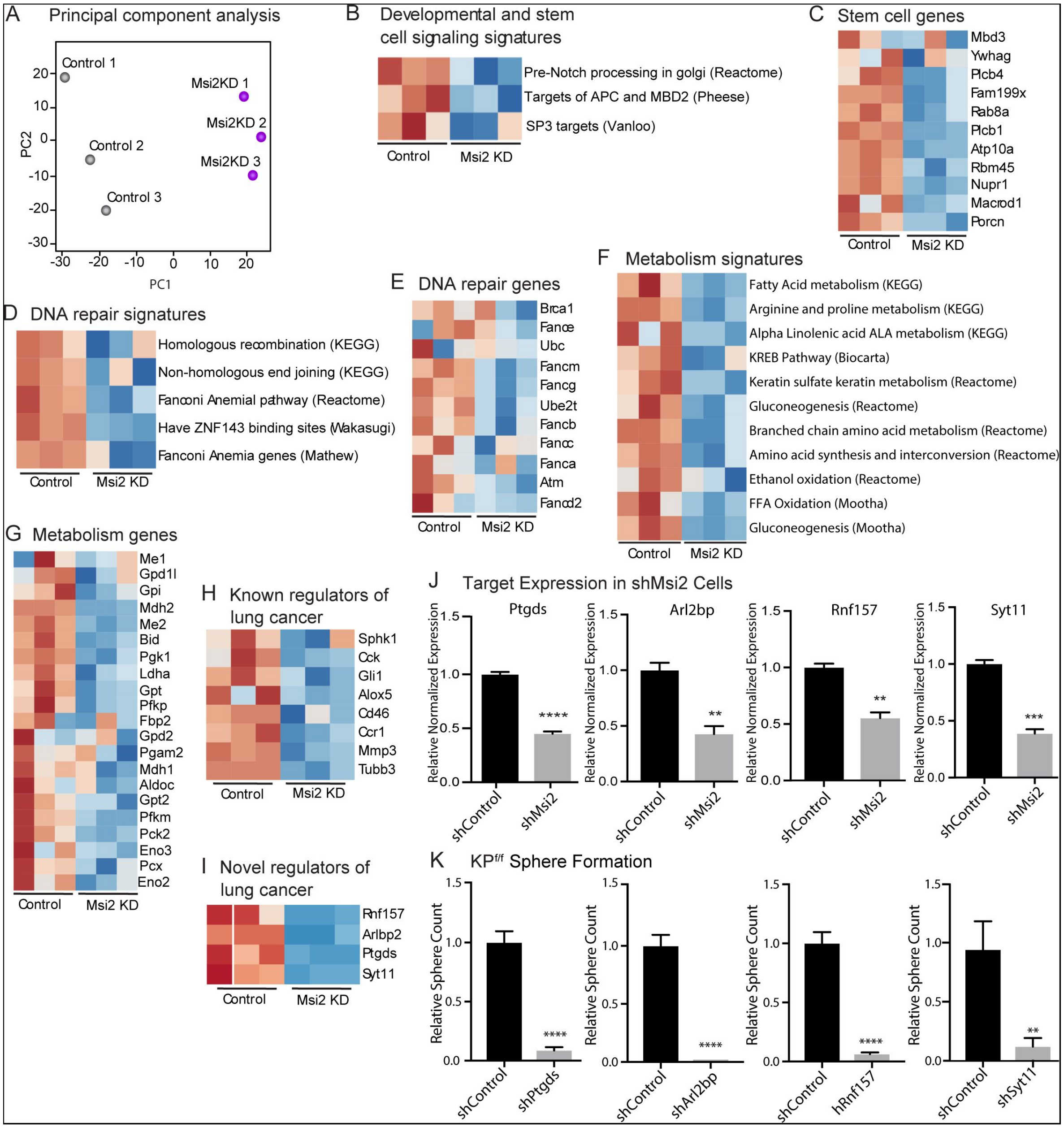


Figure 5

Msi2 regulates oncogenic signaling in lung adenocarcinoma.

A) Principal component analysis of KP cells transduced with shMsi2 (Msi2KD, purple) or shControl (gray). BG) Gene Set Enrichment Analysis of Msi2KD gene signatures. Heatmaps of gene signatures and selected genes that are involved in developmental and stem cell signaling (B-C), DNA repair (D-E), and metabolism (F-G) and are downregulated (shown in blue) following the loss of Msi2. H-I) Known (H) and novel (I) regulators of lung cancer that are downregulated following loss of Msi2. B-I) Red represents genes or gene signatures that are upregulated in the presence of Msi2, while blue represents genes that are downregulated following the loss of Msi2. J) Confirmation of downregulated genes in Msi2KD cells using qRT-PCR analysis. K) *In vitro* functional analysis of novel effectors of lung cancer (Ptgds, Arl2bp, Rnf157, Sty11) downregulated by Msi2. $KP_{f/f}$ cells were transduced with shRNA to inhibit the genes of interest and analyzed for the resulting impact on tumor sphere formation *in vitro*. Sphere formation, n=3 per condition. Data represented as mean ± SD,**p<0.01, ***p<0.001, ****p<0.0001 by Student's t-test.

adenocarcinomas, while Club cells were only able to form adenomas which did not progress further. However, when Kras^{G12D} was used in combination with p53$^{flox/flox}$, both cell types were capable of forming adenomas which could then progress to adenocarcinomas. While both genetic drivers and the cell of origin are key components for tumor formation, whether there are signaling factors that endow the cell of origin with the capacity to transform when presented with these key drivers is an area that has been less explored. In this context, our work sheds new light on such signals required at initiation as we found that Msi2 is expressed in a subpopulation of distal lung stem cells, that cells expressing Msi2 are preferentially sensitive to transformation, and that its genetic ablation leads to a marked decrease in tumor initiation and progression. The results of our RNAseq analysis suggest that this enhanced capacity may be established through an enrichment in signaling programs involved in the maintenance of an undifferentiated state as well as those known to support aggressive tumor growth which then provides a permissive environment for early tumor growth and development. These findings identify a new role for Msi2 signaling as an enhancer of tumor growth for lung adenocarcinoma and indicate that subpopulations harboring these signals may be particularly primed for transformation.

The continued dependency on Msi2 signaling following tumor initiation illustrates the potency of Msi2 as a regulator of tumor progression; thus, defining the mechanism by which Msi2 exerts this influence in lung adenocarcinoma would be critical to understanding the molecular basis of its broad influence. Here, our RNAseq analysis suggests that Msi2 may act in part as a regulator of the stem cell state. Interestingly, this analysis also showed that the role of Msi2 may extend to the regulation of processes crucial for tumor growth, such as metabolism and DNA repair, thus placing it upstream of multiple potent oncogenic signals. This work also led to the identification of several novel effectors of lung adenocarcinoma regulated by Msi2. Our functional experiments inhibiting these genes showed that loss of *Ptgds*, *Arl2bp*, *Rnf157*, and *Syt11* can block lung cancer sphere formation. Previous work has shown that Ptgds indirectly facilitates phosphorylated Sox9 nuclear translocation via activation of cAMP-dependent protein kinase A (PKA)[47] and is important in the development of Sertoli cells. While *Ptgds* has been shown previously to be expressed in human lung tumors[58] our work is the first to identify a functional role of Ptgds in lung adenocarcinoma. As a regulator of STAT3 signaling, Arl2bp directly binds to the STAT3 transcription factor, which maintains STAT3's phosphorylation status and its localization in the nucleus[48]. STAT3 activity has been implicated in NSCLC[59], thus modulating STAT3 activity through *Arl2bp* inhibition may provide an alternative therapeutic strategy. Lastly, both Rnf157 and Syt11 are predominantly expressed in the brain and have been implicated in neuronal cell survival and dendrite growth and neuronal vesicular trafficking, respectively[45,46]. Thus far, their roles have not been investigated in cancer. Our findings suggest the need for further study of these genes and raise the possibility that they could serve as future therapeutic targets.

While early treatment approaches centered principally on the use of cytotoxic agents, the treatment of lung cancer has evolved to incorporate the use of targeted therapies and immunotherapy, when appropriate, in combination with the standard cytotoxic regimen[60]. For patients with a lung cancer subtype harboring known targetable mutations, this has led to considerable improvements in the response to treatment. Yet troubling survival rates, driven largely by either late-stage detection or recurrence of the disease—even in cases for which a targeted therapy has been offered— illustrate the need for an increased understanding of the biology of lung cancer in order to inform future therapeutic approaches. The findings presented here identify a novel role for Msi2 in the growth and progression of lung adenocarcinoma and suggest that this role extends to human disease. Further, these findings identify novel effectors of lung adenocarcinoma regulated by Msi2 thereby lending new insight into the mechanism by which Msi2 exerts its influence on this disease. These new insights enhance our understanding of the biology of lung adenocarcinoma and may serve to open new avenues of focus for future therapeutic strategies.

Methods

Mice

$Msi2^{eGFP/+}$ knock-in reporter mice were generated as previously described[6] and are heterozygous for the Msi2 allele. The Msi2 mutant mouse, B6; CB-Msi2Gt(pU-21T)2Imeg (Msi2−/−) was established by gene trap mutagenesis as previously described[7]. The LSL-Kras G12D mouse, B6.129S4-Krastm4Tyj/J (stock number 008179), the p53flox/flox mouse, B6.129P2-Trp53tm1Brn/J (stock number 008462), and the *Rosa26-CreER*^{T2} mouse, B6;*129-Gt*(*ROSA*) *26Sor*$^{tm1(cre/Esr1)Tyj}$ (stock number 008463) were purchased from The Jackson Laboratory. At 9 to 16 weeks of age $5X10^6$ PFU Adeno-Cre (purchased from the University of Iowa) was delivered by intratracheal instillation as previously described[25]. Where indicated, adult mice were administered tamoxifen (Sigma) in corn oil (20 mg/mL) daily by intraperitoneal injection (150 mg per gram of body weight) for 5 consecutive days. Lungs were then isolated 3 days following the last dose of tamoxifen. Mice were bred and maintained in the animal care facilities at the University of California San Diego. All animal experiments were performed according to protocols approved by the University of California San Diego Institutional Animal Care and Use Committee.

Tissue Dissociation and FACS Analysis

For isolation of non-tumorigenic lung tissue, tissue dissociation was performed as described[61]. Briefly, mice were anesthetized, perfused with cold PBS using a peristaltic pump (Fisher Scientific), and 2mL dispase (Corning) was added to the lungs via intratracheal instillation. The lungs were then removed, placed on ice, and the lobes were minced. For isolation of lung tumors, mice were anesthetized and perfused with PBS as described, then lungs were removed and placed on ice. Visible tumors were dissected away from surrounding tissue, placed on ice in 2mL dispase, and minced. Minced non-tumorigenic lung tissue or grossly dissected lung tumors were then added to 2mg/mL collagenase/dispase in PBS (Millipore Sigma), placed in a rotisserie, and rotated for 45 minutes at 37°C. 25ng/mL DNAseI (Sigma-Aldrich) was added, and the tissue was then passed through a 100µm filter and a 40µm filter (Corning), and centrifuged at 1200rpm for 5 min at 4°C. Cells were then treated with 1mL red blood cell lysis buffer (Thermofisher) on ice for 90 seconds, washed, and resuspended in HBSS (Gibco, Life Technologies) containing 5% FBS and 2mM EDTA before staining. Analysis and cell sorting was performed using a FACSAriaIII machine (Becton Dickinson) and data were analyzed using FlowJo software (Tree Star). The following antibodies were used: rat anti-mouse EpCAM-APC (eBioscience), CD45-PE or CD45-PeCy7 (eBioscience), CD31-PE or CD31-PeCy7 (eBioscience), Sca1-PeCy7 (eBioscience). CD31 and CD45 antibodies were used to mark the Lineage$^+$ populations of the lung. Specific antibody information is summarized in **Supplementary Figure 2**.

Histology, Immunohistochemistry, and Immunofluorescence

For the isolation of tumor bearing lungs from $KP^{f/f}$ mice, mice were anesthetized, perfused with PBS as described, and 4% PFA (Fisher Scientific) was added to the lungs via intratracheal instillation. Lungs were then removed, placed in 4% PFA overnight at 4°C, and stored in 70% ethanol prior to processing. The fixed lung tissue was then paraffin embedded at the UCSD Histology and Immunohistochemistry Core at The Sanford Consortium for Regenerative Medicine according to standard protocols, and 5 µm sections were obtained. Paraffin embedded tissues were then deparaffinized in xylene and rehydrated in a 100%, 95%, 70%, 50%, and 30% ethanol series. For immunohistochemistry, endogenous peroxidase was quenched in 3% hydrogen peroxide (Fisher Scientific) for 30 min at room temperature prior to antigen retrieval. Antigen retrieval was performed for 30 min in 95–100 °C 1× citrate buffer, pH 6.0 (eBioscience). Sections were blocked in TBS or PBS containing 0.1% Triton X100 (Sigma-Aldrich), 10% goat or donkey serum (Sigma Aldrich), and 5% bovine serum albumin. For single cell suspensions of lung epithelia or lung tumors, $Lin^{-}EpCAM^{+}$ cells were isolated via FACS in the manner described previously. Cells were resuspended in DMEM:F12 (Fisher Scientific) supplemented with 50% FBS and adhered to slides by centrifugation at 500rpm. 24 hours later, cells were fixed in 4% PFA (Fisher Scientific), washed in PBS, and blocked with PBS containing 0.1% Triton X-100 (Sigma-Aldrich), 10% Goat serum (Fisher Scientific), and 5% bovine serum albumin (Invitrogen).

All incubations with primary antibodies were carried out overnight at 4°C. For immunofluorescence staining, incubation with Alexafluor-conjugated secondary antibodies (Molecular Probes) was performed for 1 hour at room temperature. DAPI (Molecular Probes) was used to detect DNA and images were obtained with a Confocal Leica TCS SP5 II (Leica Microsystems). For immunohistochemical staining, incubation with biotinylated secondary antibodies (Vector Laboratories) was performed for 45 min at 20– 25°C. Vectastain© ABC HRP Kit (Vector Laboratories) was used according to the manufacturer's protocol. Sections were counterstained with haematoxylin. The following primary antibodies were used: chicken anti-GFP (Abcam, ab13970), rabbit anti-Msi2 (Abcam, ab76148), rabbit anti-SPC (Santa Cruz, sc-13979), goat anti-CC10 (Santa Cruz, sc-9773). Specific antibody information is summarized in **Supplementary Figure 3**. Whole slide imaging was used to determine the number of tumors, tumor burden, and tumor grade in the tumor-bearing lungs of $KP^{f/f}$ mice. H&E-stained slides were scanned using Aperio AT2 digital whole slide scanning (Leica Biosystems), and tumor analysis was performed using ImageScope (Aperio). Tumor grade was scored according to the criteria described by Jackson et al., 2005[53].

Mouse Lung Cancer Cell Lines

Mouse primary lung cancer cell lines were established from end-stage $KP^{f/f}$ mice as follows: tumors were isolated and dissociated into single cell suspensions in the manner described above, then plated in 1X DMEM:F12 (Fisher Scientific) supplemented with 10%FBS, 1X pen/strep (Fisher Scientific), 1X N-2 supplement (Life Technologies), 35ug/mL bovine pituitary extract (Fisher Scientific), 20ng/mL recombinant mouse EGF (Biolegend), 20ng/mL recombinant mouse FGF (R&D Systems). Upon reaching 80% confluency the cells were collected and resuspended in HBSS (Gibco, Life Technologies) containing 5% FBS and 2mM EDTA, treated with FC block, and stained with EpCAM-APC (eBioscience), and CD31-PE (eBioscience). CD31-EpCAM+ cells were sorted and plated for at least one additional passage prior to use in functional assays. Functional assays were performed using cell lines between passage 2 and passage 6.

Patient-Derived Xenografts

Patient pleural effusate samples were obtained from Moores Cancer Center at the University of California San Diego from Institutional Review Board-approved protocols with written informed consent in accordance with the Declaration of Helsinki. Effusates were centrifuged at 1200 rpm for 5 minutes at 4°C to pellet circulating tumor cells. Cells were then treated with 1mL red blood cell lysis buffer (Thermofisher) on ice for 90 seconds, washed in HBSS (Gibco, Life Technologies) containing 5% FBS and 2mM EDTA, and resuspended in 1X DMEM:F12 (Fisher Scientific) supplemented with 10%FBS, 1X pen/strep (Fisher Scientific), 1X N-2 supplement (Life Technologies), 35ug/mL bovine pituitary extract (Fisher Scientific), 20ng/mL recombinant human EGF (Life Technologies), 20ng/mL recombinant human FGF-basic (Fisher Scientific). For each patient sample, a 1:1 ratio of resuspended cells in growth factor reduced Matrigel (Corning) at a final volume of 100 µL was transplanted into the flank NSG recipient mice to allow for the development of the patient-derived xenograft. Tumor growth was monitored via caliper measurement, and mice were euthanized when tumors reached 2 cm in diameter. Tumors were then dissociated and used for functional analysis or for subsequent passage. Functional analysis was performed using patient-derived xenografts between passage 1 and passage 2.

Lung Tumorsphere Assays

For tumorsphere assays of mouse lung cancer cell lines involving genetic inhibition via shRNA, low passage $KP^{f/f}$ cell lines were infected with shRNA lentivirus. 72-hours after transduction, the cells were sorted for positively infected cells. For tumorsphere assays involving $Msi2^{eGFP/+}$; $Kras^{G12D/+}$; $p53^{fl/fl}$; $RosaCre^{ER/+}$ mice, tamoxifen treatment and cell sorting for CD45-CD31-EpCAM+GFP+ ($Msi2^{+}$) and CD45-CD31-EpCAM+GFP-(Msi2-) cell populations were performed as described. After sorting, cells were resuspended in full media [1X DMEM:F12 (Fisher Scientific) supplemented with 10%FBS, 1X pen/strep (Fisher Scientific), 1X N-2 supplement (Life Technologies), 35ug/mL bovine pituitary extract (Fisher Scientific), 20ng/mL recombinant mouse EGF (Biolegend), 20ng/mL recombinant mouse FGF (R&D Systems)]. Resuspended cells were mixed with an equal volume of growth factor reduced Matrigel (Corning) for a total volume of 100μL, plated in 96 well flat-bottom plates (Fisher Scientific) at a density of 500 cells/well. Matrigel was allowed to solidify at 37°C for 20 min, and 150 μL of full media was added to the well. Tumorsphere cultures were incubated at 37°C for 10-14 days with the media refreshed once a week. To passage cells, tumorspheres were dissociated from surrounding Matrigel by removing the media, adding 200 μL of 2mg/mL collagenase/dispase (Millipore Sigma) to the wells, and incubating for 1 hour at 37°C. Cells were collected and placed on ice, the wells were washed 3X with full media, and each media collection was pooled on ice with the original cell collection. Cells were pelleted via centrifugation at 500 rcf for 5 minutes at 4°C, media was aspirated, 200 μL of cold Accumax (Innovative Cell Technologies) was added to the cell pellet, and cells were incubated at 37°C for 10 minutes. Cells were manually pipetted 100X to dissociate tumorspheres and examined via hemocytometer to determine if a single cell suspension had been obtained. The incubation and mechanical dissociation were repeated 2-3X until a single cell suspension was obtained. Cells were resuspended in full media, and 500 cells were plated at a 1:1 ratio with growth factor reduced Matrigel (Corning) at a final volume of 100 μL. Matrigel was allowed to solidify at 37°C for 20 min, and 150 μL of full media was added to the well, and cells were incubated in the manner described.

Flank Transplant Assays

For flank transplant assays of mouse lung cancer cell lines involving genetic inhibition via shRNA, low passage $KP^{f/f}$ cell lines were transduced with shRNA lentivirus, sorted, and 20,000 cells were resuspended in a 1:1 mix of full media and Matrigel (Corning) at a final volume of 100 μL in the manner described. Cells were injected subcutaneously into the flank of non-tumor bearing (i.e. not treated with Adeno-Cre), immunocompetent $KP^{f/f}$ mice. Subcutaneous tumors were measured with calipers once weekly for 6-15 weeks and did not exceed 2cm in diameter. At endpoint, flank tumors were removed, and tumors were dissociated as described. Tumor cells were stained with CD45-PeCy7 and CD31-PeCy7, and the absolute number of tumor cells was calculated by multiplying the percentage of CD45-CD31- transduced cells by the total number of CD45-CD31- live cells. For flank transplant assays of patient-derived xenografts, freshly dissociated xenografts were plated in full media [1X DMEM:F12 (Fisher Scientific) supplemented with 10%FBS, 1X pen/strep (Fisher Scientific), 1X N-2 supplement (Life Technologies), 35ug/mL bovine pituitary extract (Fisher Scientific), 20ng/mL recombinant human EGF (Life Technologies), 20ng/mL recombinant human FGF-basic (Fisher Scientific)]. 500,000 cells/well were plated in a 6-well plate (Fisher Scientific) coated with Matrigel (Corning) for subsequent transplant, and 83,000 cells/well were plated in a 24-well plate (Fisher Scientific) coated with Matrigel (Corning) for subsequent determination of transduction efficiency via FACS. Cells were transduced with proportionate amounts of shControl or shMsi2 lentivirus. 24 hours after transduction, media from the cells intended for transplant was collected and placed on ice. 1mL of 2mg/mL collagenase/dispase (Millipore Sigma) was then added to the well and incubated for 45 minutes at 37°C to dissociate the remaining cells from Matrigel. The wells were washed 3X with full media and each wash was collected and pooled on ice with the media previously collected. Cells were pelleted by centrifugation at 1200 rpm for 5 minutes at 4°C, an equivalent number of shControl and shMsi2 transduced cells were resuspended in full media, mixed at a 1:1 ratio with growth factor reduced Matrigel (Corning) at a final volume of 100 µL, and transplanted subcutaneously into the flanks of NSG recipient mice. Tumor growth was monitored 1-2X a week via caliper measurement for 11-12 weeks and did not exceed 2cm in diameter. 48 hours after transduction, cells from the 24-well plate were collected in the manner described using a proportionate amount of reagents, and the frequency of transduced live cells was determined via FACS. For flank transplant assays of cells from Msi2$^{eGFP/+}$; Kras$^{G12D/+}$; p53$^{fl/fl}$; RosaCre$^{ER/+}$ tumorspheres, quaternary passage tumorspheres were dissociated from surrounding Matrigel in the manner described. Cells were resuspended in full media, mixed at a 1:1 ratio with growth factor reduced Matrigel (Corning) at a final volume of 100 µL, and transplanted subcutaneously into the flanks of NSG recipient mice. Tumor growth was monitored 1-2X a week via caliper measurement for 13 weeks and did not exceed 2cm in diameter. At endpoint, both the flank tumor and lungs were collected for fixation and H&E analysis in the manner described. Lung H&E sections were analyzed for the presence of metastatic tumors.

shRNA Lentivirus Production

For knockdown of Msi2, shRNA constructs and scrambled control were designed as previously described[6]. For knockdown of Ptgds, Arl2bp, Rnf157, and Syt11, shRNA constructs and scrambled control were designed as previously described[62] using the following target sequences: For Ptgds, 5'-ACCTCTACCTTCCTCAGGAAA – 3'; for Arl2bp, 5'-GCTGCTCACATTCACGGATTT-3'; for Rnf157 5'-CAGAGGGAAATGATATCATAG – 3'; for Syt11, 5'-ATCAGGCTTCTCTGGGTTATT-3'.

qRT-PCR Analysis

RNA isolation was performed using RNeasy Micro and Mini kits (Qiagen) and cDNA conversion was performed using Superscript III (Invitrogen). Quantitative real-time PCR was performed using an iCycler (BioRad) in a mix containing iQ SYBR Green Supermix (Biorad), cDNA, and gene-specific primers. Primer sequences for target genes are shown in Supplementary Figure 3. All real-time data was normalized to B2M.

RNA sequencing and Bioinformatics Analysis

For RNA-seq of lung epithelial cells from Msi2$^{eGFP/+}$; Kras$^{G12D/+}$; p53$^{fl/fl}$; RosaCre$^{ER/+}$ mice, 60,000 cells were isolated for the Msi2^{+} (Lin-EpCAM+GFP+) and Msi2^{-}(Lin-EpCAM+GFP) populations via FACS in the manner described. For RNA-seq of Msi2 knockdown in KP$^{f/f}$ cells, KP$^{f/f}$ cells were transduced with either shControl or shMsi2 lentivirus in triplicate for each group, and ≥125,000 transduced cells were isolated via sorting. RNA was isolated using a RNeasy Micro Kit (Qiagen). The quality of total RNA was assessed using an Agilent Tapestation and all samples had a RIN ≥8. M2KD RNA libraries were generated from 2 µg of RNA, and Msi2$^{eGFP/+}$; Kras$^{G12D/+}$; p53$^{fl/fl}$; RosaCre$^{ER/+}$ libraries were generated from 90ng of RNA using Illumina's TruSeq Stranded mRNA Sample Prep Kit following the manufacturer's instructions. RNA libraries were multiplexed and sequenced with 50 basepair (bp) single end reads (SR50) to a depth of approximately 30 million reads per sample on an Illumina HiSeq4000.

RNA-seq analysis and cell state analysis were performed as previously described[62]. For validation of differential expression analysis, edgeR and sleuth were used. Briefly, RNA-seq fastq files were processed within the systemPipeR pipeline[63]. Alignment of the reads was performed using HISAT2 with the mm10 mouse assembly using default settings[64]. Read counting was performed with the summarize Overlaps R packages with Union mode and the edgeR package was used for differential expression analysis using a fold change >1.9 and an FDR <0.05[65,66]. As a secondary validation, transcript quantifications were performed using kallisto and sleuth was used for gene-level quantifications and differential expression analysis using default setting[67,68].

Statistical Analysis

Statistical analysis was performed using GraphPad Prism software version 7.0 (GraphPad Software Inc.). Data are mean ± SEM. Two-tailed unpaired Student's t-tests and Log-Rank tests were used to determine statistical significance. The Grubb's test was used to identify and remove statistical outliers where indicated.

Data Availability

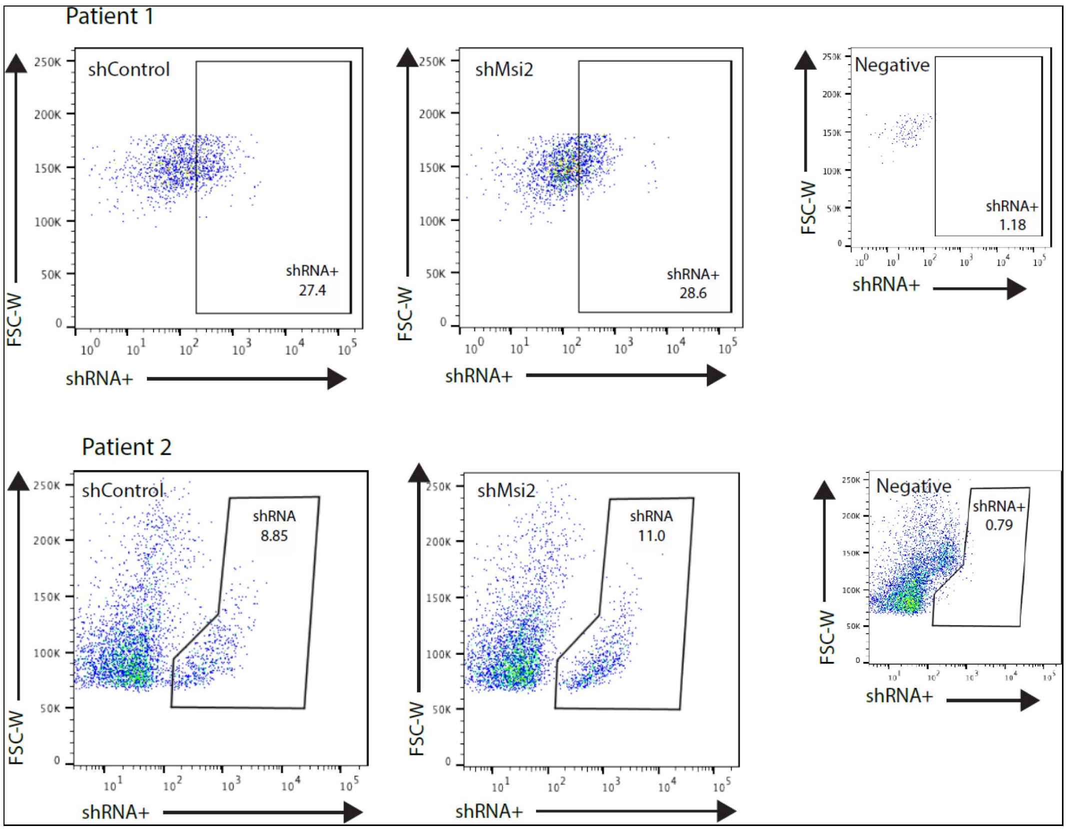
The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Acknowledgements

We are grateful to Reuben J. Shaw for scientific advice; Laurie Gerkin and Amanda Hutchins for technical training; Lillian J. Eichner for technical advice; and Marcie Kritzik for assistance with manuscript preparation.

Competing Interests

T.R. is a founder and is a member of the Scientific Advisory Board of Tiger Hill Therapeutics.



Supplementary Figure 1

FACS plots for PDXs transduced with shRNA

PDX cells transduced with either shControl or shMsi2 lentivirus have a similar frequency of infection. CD45-CD31-PDX cells from Patient 1 show a 27.4% frequency of infection when transduced with shControl, and a 28.6% frequency of infection when transduced with shMsi2 (top). CD45-CD31-PDX cells from Patient 2 show an 8.85% frequency of infection when transduced with shControl, and an 11% frequency of infection when transduced with shMsi2 (bottom). PDX cells that were untransduced and unstained for FACS antibodies were used as negative gating controls (shown at right for each patient sample).

Supplementary Figure 2

Antibodies used Tables of antibodies used in this study for FACS and immunofluorescence and immunohistochemistry

FACS Antibodies

Antibody	Fluorophore	Clone	Vendor	Dilution	Catalog Number
rat anti-mouse Epcam-APC	APC	G8.8	eBioscience	1uL/1M cells	17-5791-82
CD45-PE	PE	104	eBioscience	1uL/1M cells	12-0454-82
CD45-PeCy7	PeCy7	30-F11	eBioscience	1uL/1M cells	25-0451-82
CD31-PE	PE	390	eBioscience	1uL/1M cells	12-0311-83
CD31-PeCy7	PeCy7	390	eBioscience	1uL/1M cells	25-0311-81
Sca1-PeCy7	PeCy7	D7	eBioscience	1uL/1M cells	25-5981-82

Immunofluorescence and Immunohistochemistry

Antibody	Clone	Vendor	Dilution	Catalog Number
chicken anti-GFP	polyclonal	Abcam	1:200	ab13970
rabbit anti-Msi2	EP1305Y	Abcam	1:500	ab76148
rabbit anti-SPC	FL-197	Santa Cruz	1:100	sc-13979
goat anti-CC10	S-20	Santa Cruz	1:200	sc-9773

Supplementary Figure 3

qRT-PCR primers for genes of interest Table of qRT-PCR primers for all genes of interest as well as the housekeeping gene used in this study.

Gene	Forward primer	Reverse Primer
Msi2	5′- TGCCATACACCATGGATGCGT-3′	5′-GTAGCCTCTGCCATAGGTTGC-3′
Ptgds	5′- GAAGGCGGCCTCAATCTCA- 3′	5′-CGTACTCGTCATAGTTGGCCTC–3′
Arl2bp	5′- ACGATGGACGCCCTAGAAGA-3′	5′-GCAATAACTGGAACTCATCATCCAT-3′
Rnf157	5′- CGCATTTCTAGGCATGGAGTG-3′	5′- GCCAGGTACCGGGTAAGC-3′
Syt11	5′-GAGATCACAAATATACGCCCCAG-3′	5′-GCAGCACGTCCACACAAAG-3′
B2M	5′-ACCGGCCTGTATGCTATCCAGAA-3′	5′- AATGTGAGGCGGGTGGAACTGT-3′

References

1. American Cancer Society. (2019) **Surveillance Research**

2. Davidson M. R., Gazdar A. F., Clarke B. E (2013) **The pivotal role of pathology in the management of lung cancer** *J. Thorac. Dis* **5**

3. Langer C. J., Besse B., Gualberto A., Brambilla E., Soria J. C (2010) **The evolving role of histology in the management of advanced non - small-cell lung cancer** *J. Clin. Oncol* **28**:5311–5320

4. Heist R. S., Engelman J. A (2012) **SnapShot: Non-Small Cell Lung Cancer** *Cancer Cell* **21**:448–448

5. American Cancer Society (2024) **5-Year Survival Rates for Lung Cancer**

6. Fox R. G. *et al.* (2016) **Image-based detection and targeting of therapy resistance in pancreatic adenocarcinoma** *Nature* **534**:407–411

7. Ito T. *et al.* (2010) **Regulation of myeloid leukaemia by the cell-fate determinant Musashi** *Nature* **466**:765–768

8. Koechlein C. S. *et al.* (2016) **High-resolution imaging and computational analysis of haematopoietic cell dynamics in vivo** *Nat. Commun* **7**

9. Kudinov A. E. *et al.* (2016) **Musashi-2 (MSI2) supports TGF-β signaling and inhibits claudins to promote non-small cell lung cancer (NSCLC) metastasis** *Proc. Natl. Acad. Sci. U. S. A* **113**:6955–6960

10. Moreira A. L., Gonen M., Rekhtman N., Downey R. J (2010) **Progenitor stem cell marker expression by pulmonary carcinomas** *Mod. Pathol* **23**:889–895

11. Okano H., Imai T., Okabe M. Musashi (2002) **A translational regulator of cell fate** *J. Cell Sci.* **115**:1355–1359

12. Hope K. J., Cellot S., Ting S. B., MacRae T., Mayotte N., Iscove N. N., Sauvageau G (2010) **An RNAi screen identifies msi2 and prox1 as having opposite roles in the regulation of hematopoietic stem cell activity** *Cell Stem Cell* **7**:101–113

13. Kharas M. G. *et al.* (2010) **Musashi-2 regulates normal hematopoiesis and promotes aggressive myeloid leukemia** *Nat. Med* **16**:903–908

14. Kumar S. Srivastav *et al.* (2019) **Estrogen-dependent DLL1-mediated Notch signaling promotes luminal breast cancer** *Oncogene* **38**:2092–2107

15. Than-Trong E, Ortica-Gatti S., Mella S., Nepal C., Alunni A., Bally-Cuif L. (2018) **Neural stem cell quiescence and stemness are molecularly distinct outputs of the notch3 signalling cascade in the vertebrate adult brain** *Dev* **145**

16. Ntziachristos P., Lim J. S., Sage J., Aifantis I (2014) **From fly wings to targeted cancer therapies: A centennial for notch signaling** *Cancer Cell* **25**:318–334

17. Yang Y. *et al.* (2011) **The Notch ligand Jagged2 promotes lung adenocarcinoma metastasis through a miR-200 -Dependent pathway in mice** *J. Clin. Invest* **121**:1373–1385

18. Chin Y. R., Yoshida T., Marusyk A., Beck A. H., Polyak K., Toker A (2014) **Targeting Akt3 signaling in triple-negative breast cancer** *Cancer Res* **74**:964–973

19. Mulligan L. M (2014) **RET revisited: Expanding the oncogenic portfolio** *Nat. Rev. Cancer* **14**:173–186

20. Ramsay R. G., Gonda T. J (2008) **MYB function in normal and cancer cells** *Nat. Rev. Cancer* **8**:523–534

21. Muhammad N., Bhattacharya S., Steele R., Phillips N., Ray R. B (2017) **Involvement of c-Fos in the promotion of cancer stem-like cell properties in head and neck squamous cell carcinoma** *Clin. Cancer Res* **23**:3120–3128

22. Pinato D. J., Brown M. W., Trousil S., Aboagye E. O., Beaumont J., Zhang H., Coley H. M., Mauri F. A., Sharma R (2019) **Integrated analysis of multiple receptor tyrosine kinases identifies Axl as a therapeutic target and mediator of resistance to sorafenib in hepatocellular carcinoma** *Br. J. Cancer* **120**:512–521

23. Allocati N., Masulli M., Di Ilio C., Federici L (2018) **Glutathione transferases: Substrates, inihibitors and pro-drugs in cancer and neurodegenerative diseases** *Oncogenesis* **7**

24. Kumar S. S., Tomita Y., Wrin J., Bruhn M., Swalling A., Mohammed M., Price T. J., Hardingham J. E (2017) **High early growth response 1 (EGR1) expression correlates with resistance to anti-EGFR treatment in vitro and with poorer outcome in metastatic colorectal cancer patients treated with cetuximab** *Clin. Transl. Oncol* **19**:718–726

25. DuPage M., Dooley A. L., Jacks T (2009) **Conditional mouse lung cancer models using adenoviral or lentiviral delivery of Cre recombinase** *Nat. Protoc* **4**:1064–1072

26. Clevers H., Loh K. M., Nusse R (2014) **An integral program for tissue renewal and regeneration: Wnt signaling and stem cell control** *Science (80-.).* **346**

27. Tammela T. *et al.* (2017) **A Wnt-producing niche drives proliferative potential and progression in lung adenocarcinoma** *Nature* **545**:355–359

28. Zhou Y. *et al.* (2018) **Profiling of Stem/Progenitor Cell Regulatory Genes of the Synovial Joint by Genome-Wide RNA-Seq Analysis** *Biomed Res. Int*

29. Loughran S. J. *et al.* (2017) **Mbd3/NuRD controls lymphoid cell fate and inhibits tumorigenesis by repressing a B cell transcriptional program** *J. Exp. Med* **214**:3085–3104

30. Scully R., Chen J., Ochs R. L., Keegan K., Hoekstra M., Feunteun J., Livingston D. M (1997) **Dynamic changes of BRCA1 subnuclear location and phosphorylation state are initiated by DNA damage** *Cell* **90**:425–435

31. Kastan M., Bakkenist C (2003) **DNA damage activates ATM through intermolecular autophosphorylation and dimmer association** *Nature* **421**:499–506

32. Walden H., Deans A. J (2014) **The Fanconi Anemia DNA Repair Pathway: Structural and Functional Insights into a Complex Disorder** *Annu. Rev. Biophys* **43**:257–278

33. Leithner K. *et al.* (2015) **PCK2 activation mediates an adaptive response to glucose depletion in lung cancer** *Oncogene* **34**:1044–1050

34. Kuo C. Y., Ann D. K (2018) **When fats commit crimes: Fatty acid metabolism, cancer stemness and therapeutic resistance** *Cancer Commun* **38**:1–12

35. Jin L. *et al.* (2015) **Glutamate Dehydrogenase 1 Signals through Antioxidant Glutathione Peroxidase 1 to Regulate Redox Homeostasis and Tumor Growth** *Cancer Cell* **27**:257–270

36. Rentas S., Holzapfel N., Belew M. S., Pratt G., Voisin V., Wilhelm B. T., Bader G. D., Yeo G. W., Hope K. J (2016) **Musashi-2 attenuates AHR signalling to expand human haematopoietic stem cells** *Nature* **532**:508–511

37. Song L., Xiong H., Li J., Liao W., Wang L., Wu J., Li M (2011) **Sphingosine kinase-1 enhances resistance to apoptosis through activation of PI3K/Akt/NF-κB pathway in human non-small cell lung cancer** *Clin. Cancer Res* **17**:1839–1849

38. Han Y. *et al.* (2017) **Cholecystokinin attenuates radiation-induced lung cancer cell apoptosis by modulating p53 gene transcription** *Am. J. Transl. Res* **9**:638–646

39. Po A. *et al.* (2017) **Noncanonical GLI1 signaling promotes stemness features and in vivo growth in lung adenocarcinoma** *Oncogene* **36**:4641–4652

40. Avis I. M., Jett M., Boyle T., Vos M. D., Moody T., Treston A. M., Martínez A., Mulshine J. L (1996) **Growth control of lung cancer by interruption of 5-lipoxygenase-mediated growth factor signaling** *J. Clin. Invest* **97**:806–813

41. Wang Y., Ma L., Wang S., Bao Y., Ni C., Guan N., Zhao J., Fan X (2010) **Assessment of CAR- or CD46-Dependent adenoviral vector-mediated trail gene therapy in clinical adenocarcinoma lung cancer cells** *Oncology* **77**:366–377

42. Wang C. L., Sun B. S., Tang Y., Zhuang H. Q., Cao W. Z (2009) **CCR1 knockdown suppresses human non-small cell lung cancer cell invasion** *J. Cancer Res. Clin. Oncol* **135**:695–701

43. Stallings-Mann M. L., Waldmann J., Zhang Y., Miller E., Gauthier M. L., Visscher D. W., Downey G. P., Radisky E. S., Fields A. P., Radisky D. C (2012) **Matrix metalloproteinase induction of Rac1b, a key effector of lung cancer progression** *Sci. Transl. Med* **4**

44. McCarroll J. A. *et al.* (2015) **TUBB3/βIII-tubulin acts through the PTEN/AKT signaling axis to promote tumorigenesis and anoikis resistance in non-small cell lung cancer** *Cancer Res* **75**:415–425

45. Matz A., Lee S. J., Schwedhelm-Domeyer N., Zanini D., Holubowska A., Kannan M., Farnworth M., Jahn O., Göpfert M. C., Stegmüller J (2015) **Regulation of neuronal survival and morphology by the E3 ubiquitin ligase RNF157** *Cell Death Differ* **22**:626–642

46. Shimojo M., Madara J., Pankow S., Liu X., Yates J, Südhof T. C., Maximov A. (2019) **Synaptotagmin-11 mediates a vesicle trafficking pathway that is essential for development and synaptic plasticity** *Genes Dev.* **33**:365–376

47. Malki S., Nef S., Notarnicola C., Thevenet L., Gasca S., Méjean C., Berta P., Poulat F., Boizet-Bonhoure B (2005) **Prostaglandin D2 induces nuclear import of the sex-determining factor SOX9 via its cAMP-PKA phosphorylation** *EMBO J* **24**:1798–1809

48. Muromoto R., Sekine Y., Imoto S., Ikeda O., Okayama T., Sato N., Matsuda T (2008) **BART is essential for nuclear retention of STAT3** *Int. Immunol* **20**:395–403

49. Kharas M. G., Lengner C. J. (2017) **Stem Cells, Cancer, and MUSASHI in Blood and Guts** *Trends in Cancer* **3**:347–356

50. Uhlén M. *et al.* (2015) **Proteomics. Tissue-based map of the human proteome** *Science* **347**

51. Uhlen M. *et al.* (2010) **Towards a knowledge-based Human Protein Atlas** *Nat. Biotechnol* **28**:1248–1250

52. Wang S. *et al.* (2015) **Transformation of the intestinal epithelium by the MSI2 RNA-binding protein** *Nat. Commun* **6**

53. Jackson E. L., Olive K. P., Tuveson D. A., Bronson R., Crowley D., Brown M., Jacks T (2005) **The differential effects of mutant p53 alleles on advanced murine lung cancer** *Cancer Res* **65**:10280–10288

54. Tentler J. J., Tan A. C., Weekes C. D., Jimeno A., Leong S., Pitts T. M., Arcaroli J. J., Messersmith W. A., Eckhardt S. G (2012) **Patient-derived tumour xenografts as models for oncology drug development** *Nat. Rev. Clin. Oncol* **9**:338–350

55. Siolas D., Hannon G. J (2013) **Patient-Derived Tumor Xenografts: Transforming Clinical Samples into Mouse Models** *Cancer Res* **73**:5315–5319

56. Hidalgo M. *et al.* (2014) **Patient-Derived Xenograft Models: An Emerging Platform for Translational Cancer Research** *Cancer Discov* **4**:998–1013

57. Sutherland K. D., Song J. Y., Kwon M. C., Proost N., Zevenhoven J., Berns A (2014) **Multiple cells-of-origin of mutant K-Ras-induced mouse lung adenocarcinoma** *Proc. Natl. Acad. Sci. U. S. A* **111**:4952–4957

58. Ragolia L., Palaia T., Hall C. E., Klein J., Büyük A (2010) **Diminished lipocalin-type prostaglandin D2 synthase expression in human lung tumors** *Lung Cancer* **70**:103–109

59. Mohrherr J., Uras I. Z., Moll H. P., Casanova E (2020) **STAT3: Versatile Functions in Non-Small Cell Lung Cancer** *Cancers* **12**

60. Herbst R. S., Morgensztern D., Boshoff C (2018) **The biology and management of non-small cell lung cancer** *Nature* **553**:446–454

61. Kim C. F., Jackson E. L., Woolfenden A. E., Lawrence S., Babar I., Vogel S., Crowley D., Bronson R. T., Jacks T (2005) **Identification of bronchioalveolar stem cells in normal lung and lung cancer** *Cell* **121**:823–835

62. Lytle N. K. *et al.* (2019) **A Multiscale Map of the Stem Cell State in Pancreatic Adenocarcinoma** *Cell* **177**:572–586

63. Tyler T. W., Girke T (2016) **systemPipeR: NGS workflow and report generation environment** *BMC Bioinformatics* **17**:1–8

64. Kim D., Paggi J. M., Park C., Bennett C., Salzberg S. L (2019) **Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype** *Nat. Biotechnol* **37**:907–915

65. Lawrence M., Huber W., Pagès H., Aboyoun P., Carlson M., Gentleman R., Morgan M. T., Carey V. J. (2013) **Software for Computing and Annotating Genomic Ranges** *PLoS Comput. Biol* **9**:1–10

66. Robinson M. D., McCarthy D. J., Smyth G. K (2009) **edgeR: A Bioconductor package for differential expression analysis of digital gene expression data** *Bioinformatics* **26**:139–140

67. Bray N. L., Pimentel H., Melsted P., Pachter L (2016) **Near-optimal probabilistic RNA-seq quantification** *Nat. Biotechnol* **34**:525–527

68. Pimentel H., Bray N. L., Puente S., Melsted P., Pachter L (2017) **Differential analysis of RNA-seq incorporating quantification uncertainty** *Nat. Methods* **14**:687–690

Editors

Reviewing Editor
Haifan Lin
Yale University, New Haven, United States of America

Senior Editor
Utpal Banerjee
University of California, Los Angeles, Los Angeles, United States of America

Reviewer #1 (Public Review):

Summary:

Here, the authors, Barber AG et al, developed a new mouse model and investigated an importance of Musashi-2 in lung cancer. Specifically, they found that Musashi-2 is important for lung cancer cells as it controls cancer cell growth, and also regulates several genes that also control cancer cell growth. Development of a new Musashi-2 mouse model is a plus, which confirmed Musashi-2 importance for lung cancer survival, and finding several genes that Musashi controls that are important for lung cancer growth. Additionally, they demonstrated that Musashi-2 overexpression which is tracked by GFP is preferred in lung adenocarcinoma cells. The data is rigorous and only minor revisions are requested.

Strengths:

Authors achieved their goals, by developing new Musashi-2 mouse model, confirming Musashi-2 importance for lung cancer survival, and finding several genes that Musashi controls that are important for lung cancer growth.

Weaknesses:

The findings of Musashi-2 mouse and human lung cancer growth control are not that novel as prior publication in 2016 showed that already, again, in both human and mouse models (Kudinov et al PNAS, PMID: 27274057), and also the authors missed the point of that paper which did use both miuse and human models to show impact on inbvasion and metastasis- both in vitro and in vivo. Additionally, another publication is currently under revisions recently also generated new Musashi-2 transgenic mouse model which confirmed Musashi-2 support of lung cancer growth (Bychkov I et al, PMID: 37398283; https://www.biorxiv.org

/content/10.1101/2023.06.13.544756v1). Another weakness is that Musashi-2 cannot be effectively targeted and the new genes the authors found that Musashi-2 regulates are likely to be also difficult therapeutic targets. Therefore, impact of this new investigation is relatively modest in the field.

Major suggestions:

(1) Figure 3: it is unclear what is the efficiency of Msi2 deletion shRNA - could you demonstrate it by at least two independent methods? (QPCR, Western, or IHC?) please quantitate the data.

(2) In Figure 4, similarly, it is unclear if Msi2 depletion was effective- and what is shRNA efficiency. Please test this by at least two independent methods (QPCR, Western, or IHC) and also please quantitate the data

(3) the reason for impairment of cell growth demonstrated in Figs 3 and 4 is not clear: is it apoptosis? Necrosis? Cell cycle defects? Autophagy? Senescence? Please probe 2-3 possibilities and provide the data.

(4) Since Musashi-1 is a Musashi-2 paralogue that could compensate for Musashi-2 loss, please test Msi1 expression levels in matching Fig 3 and Fig 4 sections (in cells/ tumors with Msi2 deletion and in KP cells with Msi2 shRNA). One method could suffice here.

(5) It is not exactly clear why RNA-seq (as opposed to proteomics) was done to investigate downstream Msi2 targets (since Msi2 is in first place, translational and not transcriptional regulator)- . RNA effects in Fig 5J are quite modest, 2-fold or so. It would be useful (if antibodies available) to test four targets in Fig 5J by Western blot, to see any impact of musashi-2 depletion on those target protein levels. Indeed, several papers - including Kudinov et al PNAS, PMID: 27274057, Makhov P et al PMID: 33723247 and PMID: 37173995 - used proteomics/ RIP approaches and found direct Musashi-2 targets in lung cancer, including EGFR, and others.

https://doi.org/10.7554/eLife.97021.1.sa3

Reviewer #2 (Public Review):

Summary:

Alison G. Barber et al. reports the function of Msi2 in mouse models of non-small cell lung cancer. The expression of Msi2 in normal lung was evaluated using a knockin reporter allele. Msi2 expressing cells were found to be around 30-40% in normal lung epithelium without a strong bias in subsets of lung cells. Knocking out Msi2 in a KrasG12D and P53 KO model reduced lung cancer initiation. Knocking down Msi2 in established lung cancer cells reduced in vitro sphere formation and in vivo xenograft. Finally, the authors identified several genes whose expression was downregulated by Msi2 knockdown. Knocking down four of these genes, including Ptgds, Arl2bp, hRnf157, and Syt11, each with a single shRNA, reduced lung sphere formation in vitro, suggesting their involvement in lung cancer.

Strengths:

This manuscript represents an interesting advance on the role of Msi2 in lung cancer. While some of the data (for example the knockdown effect of Msi2 in established lung cancer cells) corroborated previous findings, the study of Msi2 expression in normal lung and the characterization of the KO phenotype in lung cancer initiation are new and interesting.

Weaknesses:

Two areas can be further strengthened. Several conclusions are not fully supported by the existing data. The stable/dynamic nature of Msi2 expressing cells in lung would benefit from more detailed investigations for proper data interpretation.

(1) It will be interesting to determine whether Msi2+ cells are a relatively stable subset or rather the Msi2+ cells in lung is a dynamic concept that is transient or interconvertible. This is relevant to the interpretation of what Msi2 positivity really means.

(2) Does Kras mutation and/or p53 loss upregulate Msi2? This point and the point above are related to whether Msi2+ cells are truly more susceptible to tumorigenesis, as the authors suggested.

(3) The KO of Msi2 reducing tumor number and burden in the lung cancer initiation model is interesting. However, there are two alternative interpretations. First, it is possible that the Msi2 KO mice (without Kras activation and p53 loss) has reduced total lung cell numbers or altered percentage of stem cells. There is currently only one sentence citing data not shown on line 125, commenting that there is no difference in BASC and AT2 cell populations. It will be helpful that such data are shown and the effect of KO on overall lung mass or cellularity is clarified. Second, the phenotype may also be due to a difference in the efficiencies of cre on Kras and p53 in the Msi2 WT and KO mice.

(4) All shRNA experiments (for both Msi2 KD and the KD of candidate genes) utilized a single shRNA. This approach cannot exclude off-target effects of the shRNA.

(5) The technical details of the PDX experiment (Figure 4F) are not fully explained.

https://doi.org/10.7554/eLife.97021.1.sa2

Reviewer #3 (Public Review):

Summary:

In this manuscript, Barber and colleagues propose a dual role for the RNA-binding protein Mushashi-2 (Msi2) in lung adenocarcinoma initial transformation and subsequent tumor propagation. First, authors show that Msi2 is expressed in a subset of Club/BASC (37%) and AT2 (26%) cells in the normal lung and displayed a distinct transcriptional profile than non-expressing Msi2 cells. Furthermore, Msi2 is broadly expressed/activated in vivo in genetically induced lung adenocarcinoma tumors (Kras/p53 mouse model) and Msi2+ cells displayed a significantly higher ability to form tumor spheres in vitro. Authors demonstrated by in vivo and in vitro assays that Msi2 loss of function significantly impair tumor growth and progression in lung adenocarcinoma. Data showed that Msi2 function is conserved in human adenocarcinoma tumor growth in patient-derived xenograft. Lastly, novel genes regulated by Msi2 and involved in lung adenocarcinoma tumor growth were identified.

Strengths:

The authors provided convincing data for a key role of Msi2 in lung adenocarcinoma tumor progression and growth. Multiple evidences using Msi2 knock-out genetic mouse model and shRNA knock-down in tumor sphere formation assay are clearly demonstrated. The conservation and importance of Msi2 was further shown in human patient-derived xenograft. Although specific cell types (Club/BASC, AT2) were not isolated, authors further delved in the transcriptional difference between Msi2+ and Msi2- cells in the normal lung. Furthermore, novel genes and pathways regulated by Msi2 in lung adenocarcinoma were identified and tested for their ability to inhibit tumor growth in vitro. These 2 RNA-Seq datasets will be useful in the future and provide a basis to explore 1) potential propensity of a given cell to initiate oncogenic transformation, and 2) potential novel regulators of lung adenocarcinoma.

Weaknesses:

Although this work strongly demonstrated the importance of Msi2 in lung adenocarcinoma tumor progression and growth, the following points remain to be clarified or addressed.

- In Figure 1, characterization of Msi2 expression in the normal mouse lung was carried out by using a Msi2-GFP Knock-in reporter and analyzed by flow cytometry followed by cytospins and immunostaining. Additional characterization of Msi2 expression by co-immunostaining with well-known markers of airway and alveolar cell types in intact lung tissue will strengthen the existing data and provide more specific information about Msi2 expression and abundancy in relevant cell types. It will be also interesting to know whether Msi2 is expressed or not in other abundant lung cell types such as ciliated and AT1 cells.

- While this set of experiments provide strong evidence that Msi2 is required for tumor progression and growth in lung adenocarcinoma, it is unclear whether normal Msi2+ lung cells are more responsive to transformation or whether Msi2 is upregulated early during the process of tumorigenesis. Future lineage tracing experiments using Msi2-CreER and mouse models of chemically-induced lung carcinogenesis will provide additional data that will fully support this claim.

- In Figure 4F, Patient-derived xenograft (PDX) assays were conducted in 2 patients only and the percentage of cells infected by shRNA-Msi2 is low in both PDX (30% and 10% for patient 1 and 2 respectively). It is surprising that Msi2 downregulation in a small percentage of tumor cells has such a dramatic effect on tumor growth and expansion. Confirmation of this finding with additional patient samples would suggest an important non-cell autonomous role for Msi2 in lung adenocarcinoma.

https://doi.org/10.7554/eLife.97021.1.sa1

Author response:

> ***Reviewer #1 (Public Review):***
>
> *(1) Figure 3: it is unclear what is the efficiency of Msi2 deletion shRNA - could you demonstrate it by at least two independent methods? (QPCR, Western, or IHC?) please quantitate the data.*

In Figure 3, we did not delete Msi2 via shRNA. Instead, we utilized a genetic model in which the Msi2 gene was disrupted via gene trap mutagenesis. We have also used this model in previous publications to define the impact of Msi2 loss in other systems1.

> *(2) In Figure 4, similarly, it is unclear if Msi2 depletion was effective- and what is shRNA efficiency. Please test this by at least two independent methods (QPCR, Western, or IHC) and also please quantitate the data*

We demonstrated that the efficiency of Msi2 depletion was ~83% (Figures 4A and 4C) via qPCR analysis for our in vitro and in vivo experiments, respectively, and verified the knockdown via bulk RNA-seq analysis. The shRNA hairpin used was previously validated and published by our lab2.

> *(3) the reason for impairment of cell growth demonstrated in Figs 3 and 4 is not clear: is it apoptosis? Necrosis? Cell cycle defects? Autophagy? Senescence? Please probe 2-3*

> *possibilities and provide the data.*

The basis of the cell growth impairment after Msi2 deletion/knockdown in this paper is certainly an important question, and future experiments will be performed to better delineate this. In previous publications loss of Msi2 in leukemia cells has been shown to inhibit growth via arrested cell cycle progression by increasing the expression of p213. Further, loss of Msi2 was also shown to promote apoptosis in part by upregulating Bax3. These data suggest that Msi2 can have an impact via multiple distinct mechanisms including by mediating cell cycle arrest and blocking apoptosis. While these specific genes were not detectably changed after loss of Msi2 in lung cancer cells, other genes in these and other pathways will be important to study in the future.

> *(4) Since Musashi-1 is a Musashi-2 paralogue that could compensate for Musashi-2 loss, please test Msi1 expression levels in matching Fig 3 and Fig 4 sections (in cells/ tumors with Msi2 deletion and in KP cells with Msi2 shRNA). One method could suffice here.*

In our RNA-seq of cells following Msi2 knockdown, Msi1 expression was undetectable. The TPM values for Msi1 in control and knockdown cells were less than 0.01, suggesting that it did not compensate for the loss of Msi2.

> *(5) It is not exactly clear why RNA-seq (as opposed to proteomics) was done to investigate downstream Msi2 targets (since Msi2 is in first place, translational and not transcriptional regulator)- . RNA effects in Fig 5J are quite modest, 2-fold or so. It would be useful (if antibodies available) to test four targets in Fig 5J by Western blot, to see any impact of musashi-2 depletion on those target protein levels. Indeed, several papers - including Kudinov et al PNAS, PMID: 27274057, Makhov P et al PMID: 33723247 and PMID: 37173995 - used proteomics/ RIP approaches and found direct Musashi-2 targets in lung cancer, including EGFR, and others.*

Previous published work from the lab showed that expression of Msi2 in the context of myeloid leukemia1can not only repress NUMB protein (I believe protein should be all caps?) (as has been previously demonstrated in the nervous system) but also Numb RNA. This indicated that as an RNA binding protein, Msi2 also can bind and destabilize direct binding targets such as Numb; this was the reason for pursuing transcriptomic analysis. However as the reviewer suggests, proteomic studies are certainly very important to develop a complete picture of the impact of Musashi to determine which targets are controlled by Msi2 at the protein level.

> ***Reviewer #2 (Public Review):***
>
> *(1) It will be interesting to determine whether Msi2+ cells are a relatively stable subset or rather the Msi2+ cells in lung is a dynamic concept that is transient or interconvertible. This is relevant to the interpretation of what Msi2 positivity really means.*

In previous unpublished work from our lab, we have found that Msi2+ cells from a GFP reporter KPf/fC mouse are readily able to become GFP negative (Msi2-), but the inverse is not true. Specifically, when Msi2+ KPf/fC pancreatic cells were transplanted into the flanks of NSG mice, Msi2+ cells formed tumors in all recipients; these tumors contained both GFP+ and GFP- cells (over 80%) recapitulating the original heterogeneity and suggesting GFP+ cells can give rise to both GFP+ and GFP- cells (Lytle and Reya, unpublished observations). In contrast only a small subset of GFP- transplanted mice formed tumors. One of the rare GFP- derived tumors was isolated and found to contain largely GFP- cells, with ~0.1% GFP+ cells. The small frequency of GFP expression could be from contaminating cells or may suggest that GFP- cells retain some ability to switch on Msi under selective pressure, and that although they pose a lower risk of driving tumorigenesis than Msi+ cells, they may nonetheless bear latent potential to become higher risk. These data may offer a possible model for projecting the potential of Msi2+ cells in the lung, but is something that needs to be further studied in this tissue.

> *(2) Does Kras mutation and/or p53 loss upregulate Msi2? This point and the point above are related to whether Msi2+ cells are truly more susceptible to tumorigenesis, as the authors suggested.*

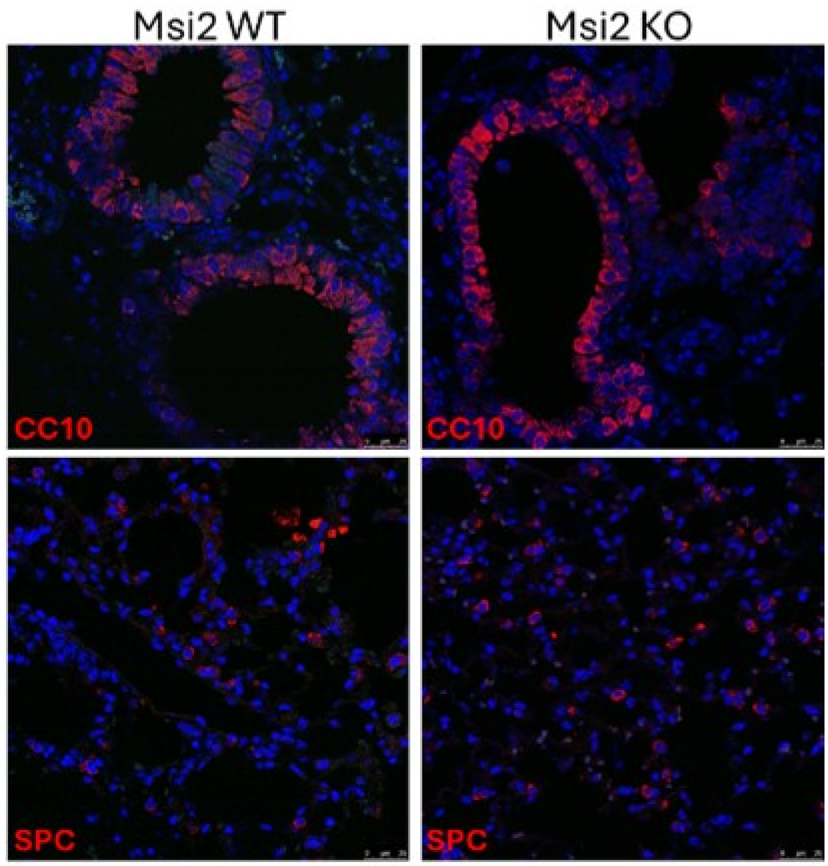
In unpublished work from our lab, we have found that Kras mutation upregulates Msi2 over baseline and subsequent p53 loss upregulates Msi2 further in the context of pancreatic cells (Lytle and Reya unpublished results), therefore it is possible that the same is true for the lung. Specifically, we have observed that Msi2 increased from normal acinar cells to Kras-mutated acinar (e.g. pancreatic intraepithelial neoplasia (PanIN)).

To address whether Msi2+ cells are more susceptible to tumorigenesis, we have recently published data showing that the stabilization of the oncogenic MYC protein in lung Msi2+ cells drive the formation of small-cell lung cancer in a new inducible Msi2-CreERT2; CAG-LSL-MycT58A mice (Msi2-Myc)4 model. More importantly, this data provides the first evidence that normal Msi2+ cells are primed and highly sensitive to MYC-driven transformation across many organs and not just the lung4.

> *(3) The KO of Msi2 reducing tumor number and burden in the lung cancer initiation model is interesting. However, there are two alternative interpretations. First, it is possible that the Msi2 KO mice (without Kras activation and p53 loss) has reduced total lung cell numbers or altered percentage of stem cells. There is currently only one sentence citing data not shown on line 125, commenting that there is no difference in BASC and AT2 cell populations. It will be helpful that such data are shown and the effect of KO on overall lung mass or cellularity is clarified. Second, the phenotype may also be due to a difference in the efficiencies of cre on Kras and p53 in the Msi2 WT and KO mice.*

We isolated the lungs of three Msi2 WT and three Msi2 KO mice and used immunofluorescence staining to stain for CC10 (BASC) and SPC (AT2) to determine if these cell populations were reduced after Msi2 loss alone. Below are representative images showing that the Msi2 KO mice did not have lower numbers of both BASC and AT2 cell populations.

Author response image 1.



> *(4) All shRNA experiments (for both Msi2 KD and the KD of candidate genes) utilized a single shRNA. This approach cannot exclude off-target effects of the shRNA.*

The shRNA hairpin used for Msi2 was previously validated and published by our lab2. Additionally, in this work we did develop and use a Msi2 genetic knockout mouse model that validates our shRNA knockdown data showing the specific impact of Msi2 on lung tumor growth.

> *(5) The technical details of the PDX experiment (Figure 4F) are not fully explained.*

Due to space considerations, we were unable not put the specifics in the legend, but the details are in the methods section (Flank Transplant Assays). In brief, 500,000 cells/well were plated in a 6-well plate coated with Matrigel and 83,000 cells/well were plated in a 24-well plate coated with Matrigel for subsequent determination of transduction efficiency via FACS. 24 hours after transduction, media from the cells was collected and placed on ice. 1mL of 2mg/mL collagenase/dispase was then added to the well and incubated for 45 minutes at 37ºC to dissociate the remaining cells from Matrigel followed by subsequent washes. Cells were pelleted by centrifugation and an equivalent number of shControl and shMsi2 transduced cells were resuspended in full media, mixed at a 1:1 ratio with growth factor reduced Matrigel at a final volume of 100 µL, and transplanted subcutaneously into the flanks of NSG recipient mice.

> ***Reviewer #3 (Public Review):***
>
> *- In Figure 1, characterization of Msi2 expression in the normal mouse lung was carried out by using a Msi2-GFP Knock-in reporter and analyzed by flow cytometry followed by cytospins and immunostaining. Additional characterization of Msi2 expression by co-immunostaining with well-known markers of airway and alveolar cell types in intact lung tissue will strengthen the existing data and provide more specific information about Msi2 expression and abundancy in relevant cell types. It will be also interesting to know whether Msi2 is expressed or not in other abundant lung cell types such as ciliated and AT1 cells.*

We performed co-staining of Msi2 and CC10 as well as Msi2 and SPC in Figure 1C. In the future we can include additional markers as well as markers for airway and other alveolar cell types.

> *- While this set of experiments provide strong evidence that Msi2 is required for tumor progression and growth in lung adenocarcinoma, it is unclear whether normal Msi2+ lung cells are more responsive to transformation or whether Msi2 is upregulated early during the process of tumorigenesis. Future lineage tracing experiments using Msi2-CreER and mouse models of chemically-induced lung carcinogenesis will provide additional data that will fully support this claim.*

Recently, we published data showing that Msi2 is expressed in Clara cells at the bronchoalveolar junction in the lung of our new Msi2-CreERT2 knock-in mouse model4. Furthermore, stabilization of the oncogenic MYC protein in these specific cells to model Myc amplification was sufficient to drive the formation of small-cell lung cancer4. These data excitingly demonstrate that Msi2+ cells are more responsive to transformation after Myc stabilization.

> *- In Figure 4F, Patient-derived xenograft (PDX) assays were conducted in 2 patients only and the percentage of cells infected by shRNA-Msi2 is low in both PDX (30% and 10% for patient 1 and 2 respectively). It is surprising that Msi2 downregulation in a small percentage of tumor cells has such a dramatic effect on tumor growth and expansion. Confirmation of this finding with additional patient samples would suggest an important non-cell autonomous role for Msi2 in lung adenocarcinoma.*

In the future we hope to collect more patient samples to further validate the data presented with the first 2 patients shown here. We are not certain about the reason behind the large impact of Msi2 inhibition, but as cancer stem cells drive the formation of the rest of the tumor and also drive the stromal microenvironment, it is possible that when Msi2 is deleted, Msi2- cells no longer form tumors? and also the ability to build the stromal microenvironment is impacted. This possibility needs to be further tested in future experiments.

References

(1) Ito, T. Kwon, H. Y., Zimdahl, B., Congdon, K. L., Blum, J., Lento, W. E., Zhao, C., Lagoo, A., Gerrard, G., Foroni, L., Goldman, J., Goh, H., Kim, S. H., Kim, D. W., Chuah, C., Oehler, V. G., Radich, J. P., Jordan, C. T., & Reya, T. Regulation of myeloid leukaemia by the cell-fate determinant Musashi. Nature 466, 765–768 (2010).

(2) Fox, R. G. Lytle, N. K., Jaquish, D. V., Park, F. D., Ito, T., Bajaj, J., Koechlein, C. S., Zimdahl, B., Yano, M., Kopp, J. L., Kritzik, M., Sicklick, J. K., Sander, M., Grandgenett, P. M., Hollingsworth, M. A., Shibata, S., Pizzo, D., Valasek, M. A., Sasik, R., Scadeng, M., Okano, H., Kim, Y., MacLeod,

A. R., Lowy, A. M., & Reya, T. Image-based detection and targeting of therapy resistance in pancreatic adenocarcinoma. Nature 534, 407–411 (2016).

(3) Zhang, H. Tan, S., Wang, J., Chen, S., Quan, J., Xian, J., Zhang, Ss., He, J., & Zhang, L. Musashi2 modulates K562 leukemic cell proliferation and apoptosis involving the MAPK pathway. Exp Cell Res 320, 119-27 (2014).

(4) Rajbhandari, N., Hamilton, M., Quintero, C.M., Ferguson, L.P., Fox, R., Schürch, C.M., Wang, J., Nakamura, M., Lytle, N.K., McDermott, M., Diaz, E., Pettit, H., Kritzik, M., Han, H., Cridebring, D., Wen, K.W., Tsai, S., Goggins, M.G., Lowy, A.M., Wechsler-Reya, R.J., Von Hoff, D.D., Newman, A.M., & Reya, T. Single-cell mapping identifies MSI+ cells as a common origin for diverse subtypes of pancreatic cancer. Cancer Cell 41(11):1989-2005.e9 (2023).

https://doi.org/10.7554/eLife.97021.1.sa0